

Embedded Systems and IoT

Ingegneria e Scienze Informatiche - UNIBO

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[module-3.3]

COMMUNICATION TECHNOLOGIES AND PROTOCOLS IN IOT AN OVERVIEW

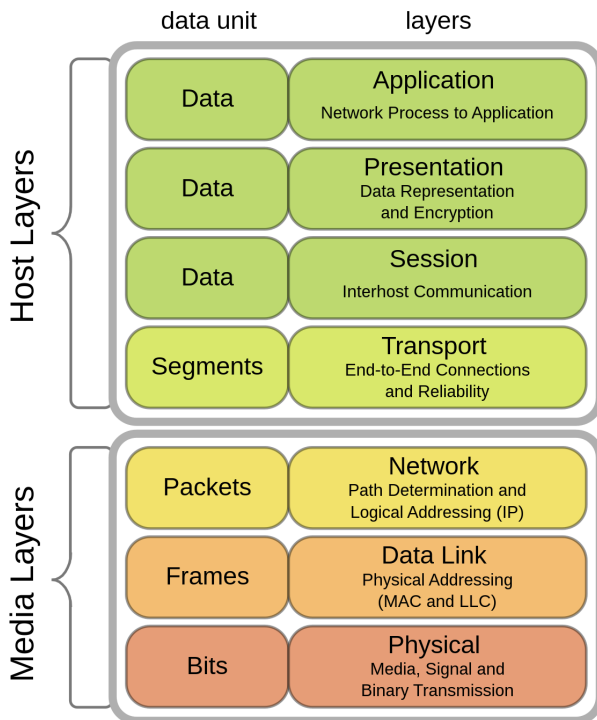
COMMUNICATION IN M2M SYSTEMS AND INTERNET OF THINGS

- Communication among devices is a crucial aspect of M2M (Machine-to-Machine) and IoT systems, that is the capability to exchange data/information among devices and subsystems
- Communication can be either **wired** or **wireless**.
- In both cases devices are equipped with suitable communication modules
 - in the *wired* case, these are shields/boards with an Ethernet port and support for 802.11 protocol
 - in the *wireless* case, they could be either shield or USB dongle or event SoC including a transmitter/receiver module and the antenna
- The communication between the microcontroller and the communication modules typically occurs through the serial port

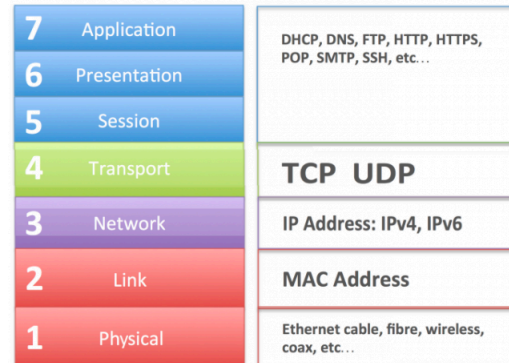
INTERNET CONNECTION

- There are 2 main ways to connect an embedded device to the Internet network in IoT:
 - **indirectly**, by means of the communication with an intermediary node functioning as a **gateway**
 - e.g. architectures based on ZigBee or Bluetooth wireless technology
 - **directly**, mounting a proper receiver/transmitter module
 - e.g. 3G/4G/5G or wifi
- In both cases, the reference protocol is the Internet protocol
 - IP, TCP/IP and UDP/IP

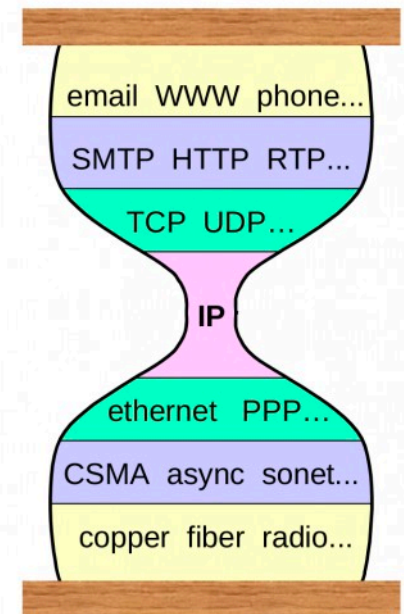
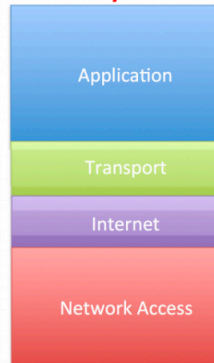
ISO-OSI ARCHITECTURE & INTERNET STACK/“HOURLGLASS”



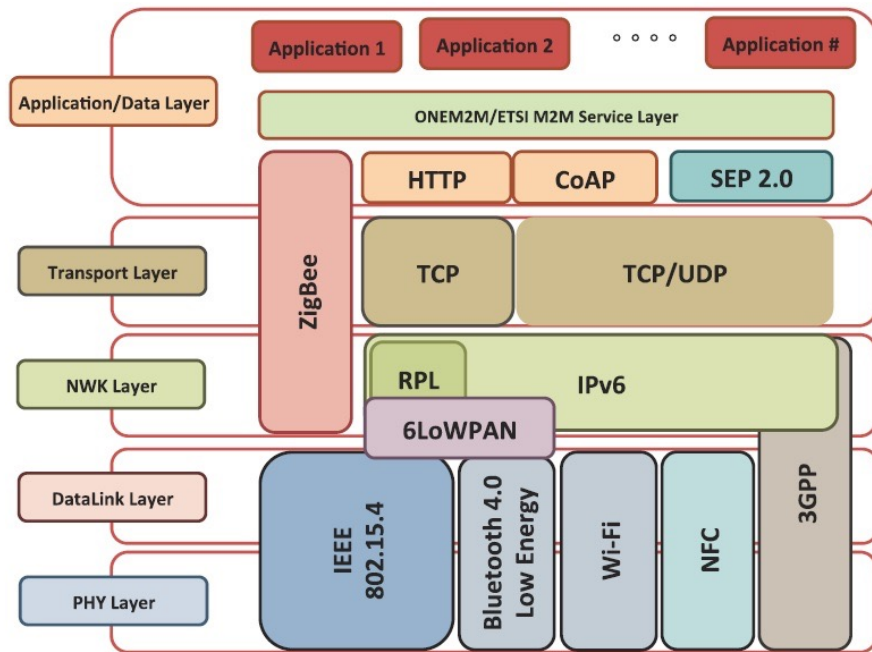
The OSI Model



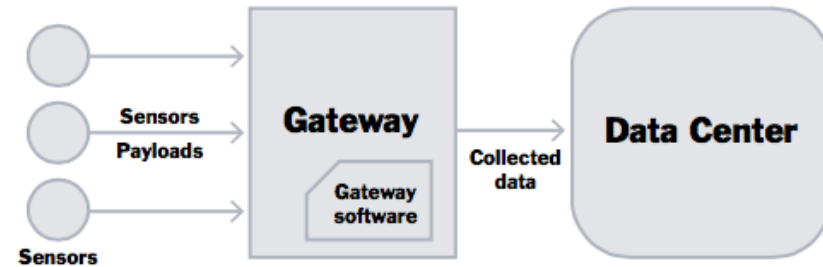
The TCP/IP Model



IOT PERSPECTIVE



Heterogeneous standards environment in IoT (Source: IoT Research EU)

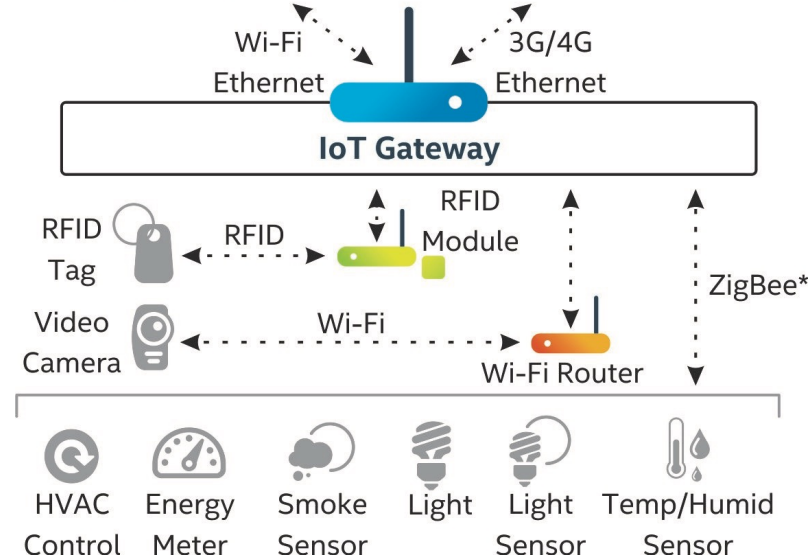


User Interface Devices

Control and Monitoring



Tablet, Smart Watch, etc.



<https://blogs.intel.com/iot/2015/04/15/how-the-internet-of-things-can-unlock-the-door-to-a-more-robust-bms/>

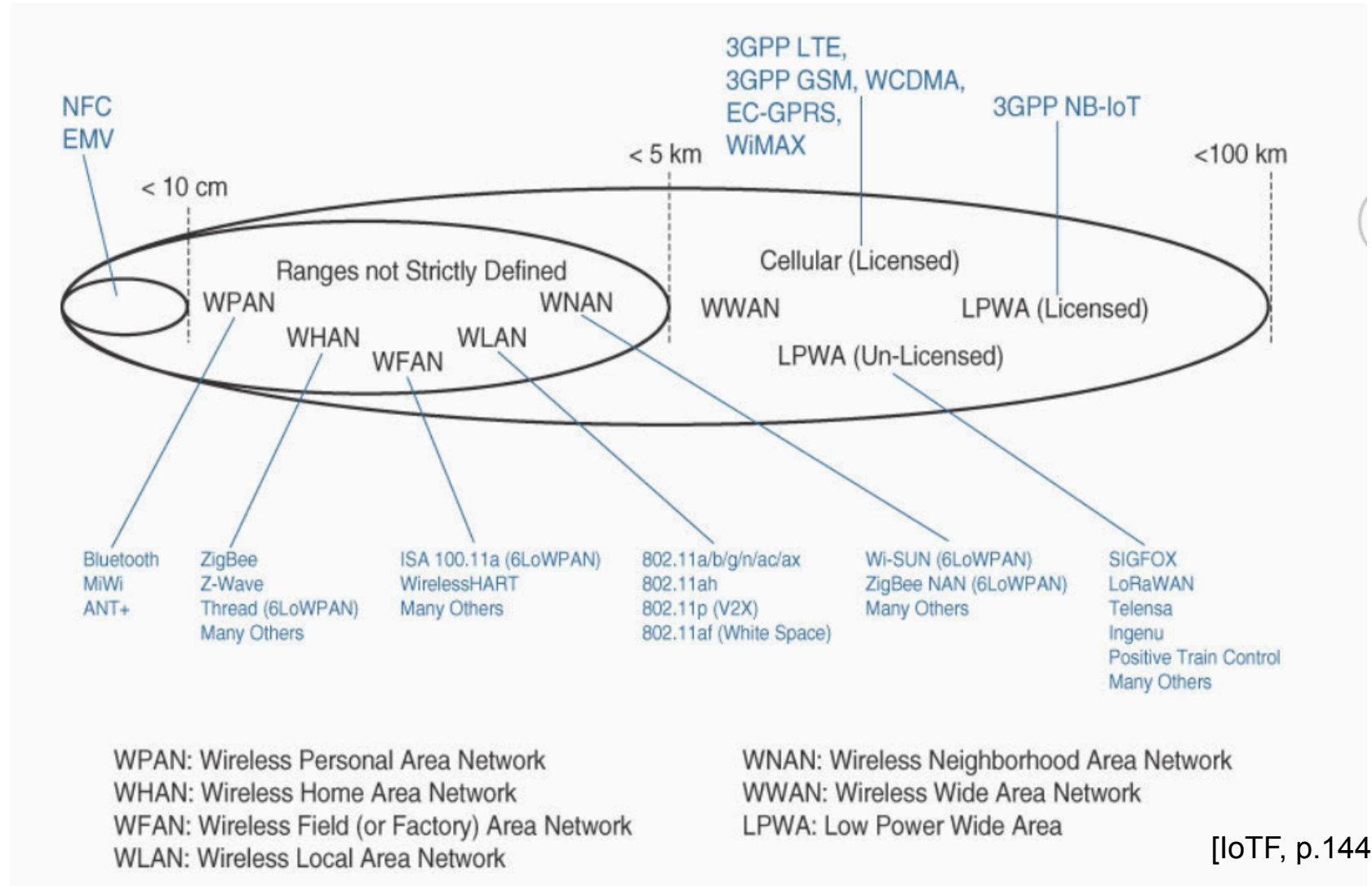
WIRELESS COMMUNICATION: PREAMBLE

- Wireless communications are **radio communications**
 - exploiting electromagnetic signals belonging to the radio frequencies / bandwidth, enabling the transmission of information at distance via air
 - signals sent on these frequencies are also called radio-frequency signals (RF)
- A radio communication can be
 - *terrestrial* — if based on telecom infrastructure located on the earth's surface
 - *satellite* — if based on infrastructures in orbit over Earth

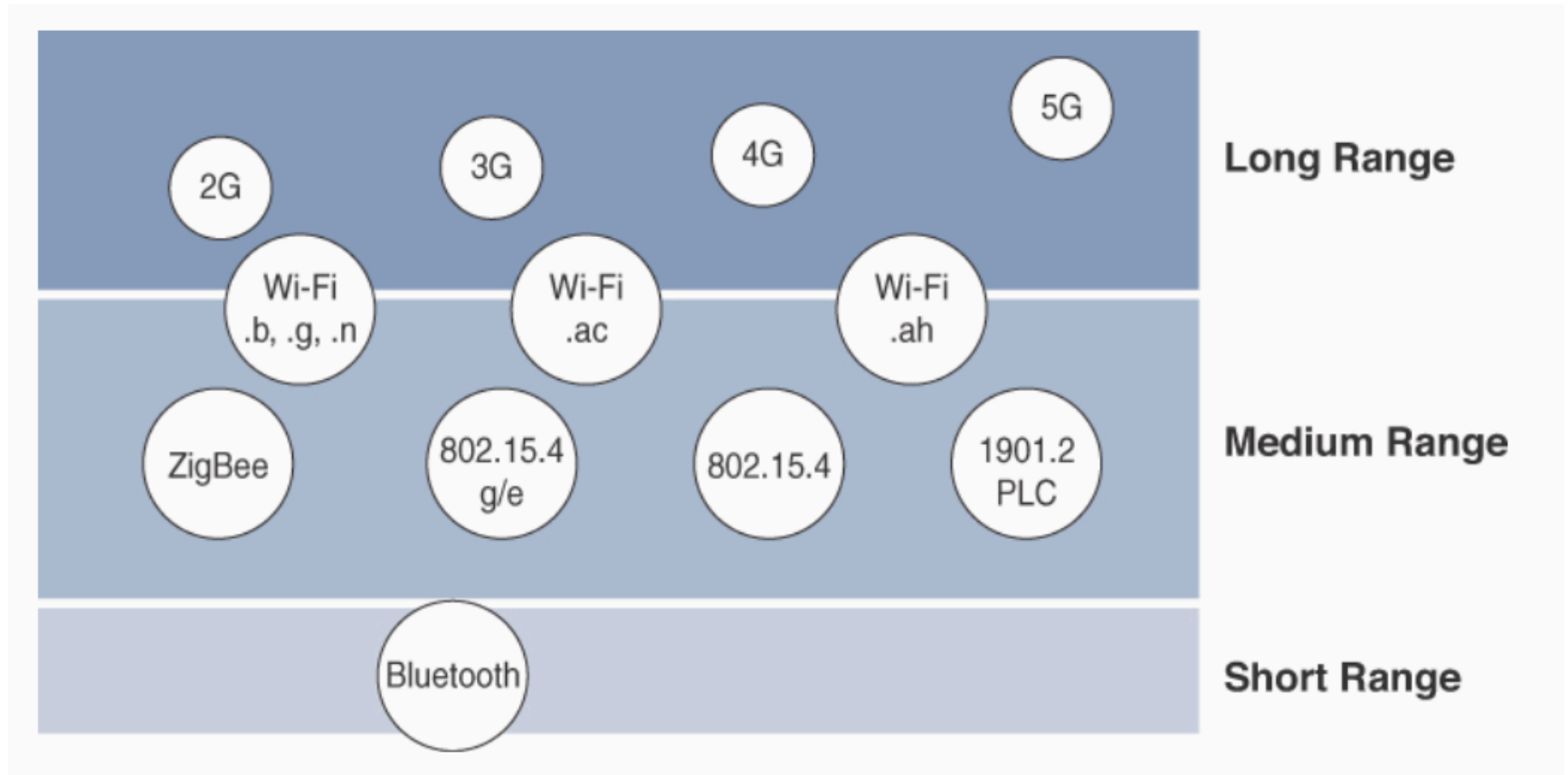
WIRELESS TECHNOLOGIES AND PROTOCOLS

- **Short Range** (~ten meters)
 - consumer field, less used in industry
 - examples
 - IEEE 802.15.1 (Bluetooth)
 - IEEE 802.15.7 (Visible Light Communication)
- **Medium Range** (hundred meters, up to one mile)
 - examples
 - IEEE 802.11 Wi-Fi
 - IEEE 802.15.4 e IEEE 802.15.4g WPAN
- **Long Range** (distances > 1 mile)
 - cellular network (2G, 3G, 4G, **5G**)
 - outdoor IEEE 802.11 Wi-Fi
 - Low-Power Wide-Area (LPWA) technologies

WIRELESS TECHNOLOGIES AND PROTOCOLS



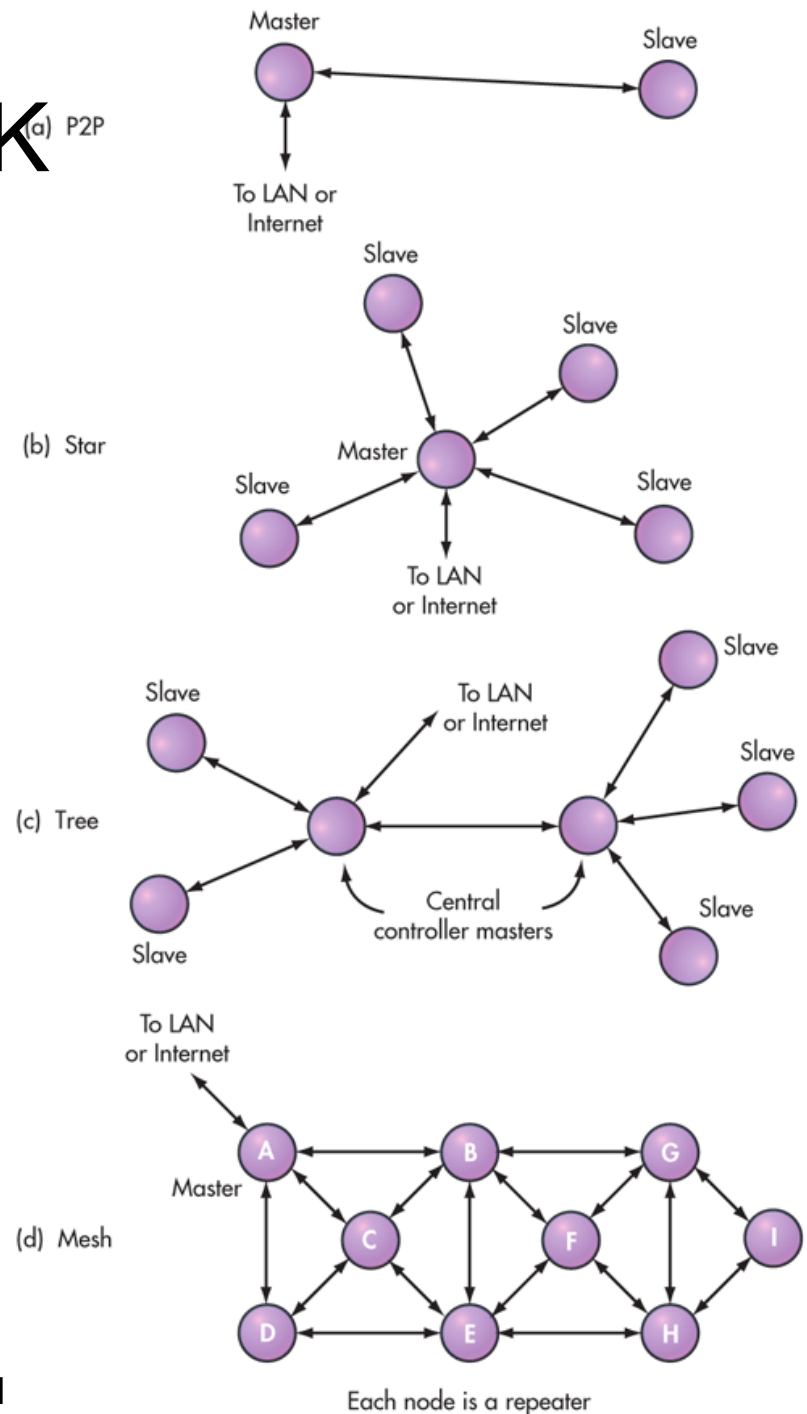
WIRELESS TECHNOLOGIES AND PROTOCOLS



[IoT, p.230]

WIRELESS NETWORK TOPOLOGIES

- Main topologies used in embedded and IoT
 - P2P
 - Star
 - Tree
 - Mesh



WIFI

- Standard IEEE 802.11x
- Reference protocol for wireless connection of devices to the Internet
 - support for Internet protocols
 - IP, TCP/IP, UDP/IP
- Characteristics
 - Data rate: 54 Mbps
 - Communication range: ~150 m
 - Frequency: 5 GHz
- Used also in embedded
 - ... even if not designed for embedded systems
 - high power consumption

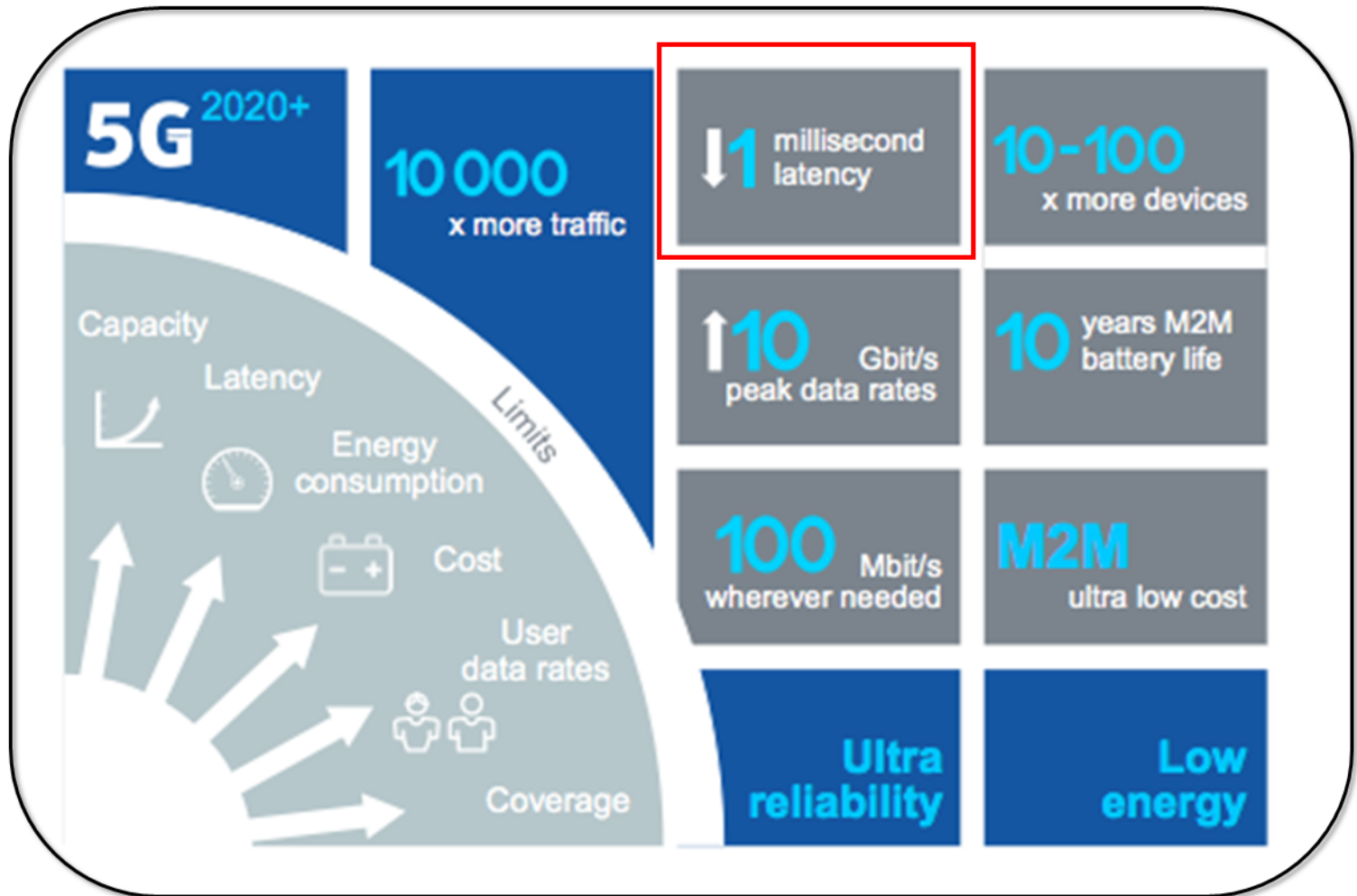
CELLULAR NETWORKS (BEFORE 5G)

- Communication by means of broad-band data cellular network
- Continuous evolution
 - GPRS, 3G, WiMax, LTE (4G)
- Characteristics
 - Data rate: from 80 Kbps (GPRS) to some Mbps (3G e 4G)
 - Range: km
 - Frequencies: 800 MHz, 1900 MHz
- High power consumption
 - not ideal for IoT

5G

- Disruptive technology, designed (also for) IoT
 - high data rate/speed: 1-10Gbps
 - low latencies (1-10ms)
 - greater reliability
 - greater flexibility compared to Wifi, pervasiveness in devices (sensors, wearable..)
- introduzione/overview
 - <https://www.cbinsights.com/research/5g-next-gen-wireless-system/>
 - https://www.arpae.it/dettaglio_generale.asp?id=4149&idlivello=2145

5G AND IOT

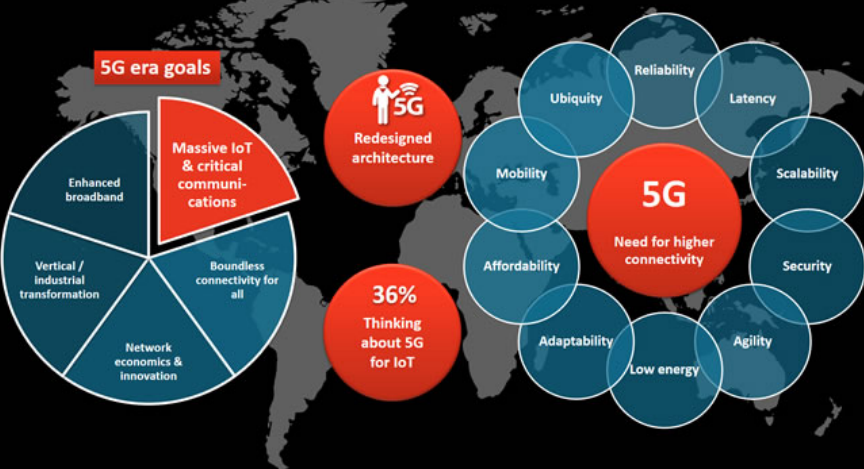


THE FUTURE OF 5G AND IOT
Finishing the foundations in 2018 and 2019 for the next two decades

www.i-scoop.eu/internet-of-things-guide/5g-iot/

By 2021, 5G's broad enablement of IoT use cases will drive 70% of G2000 companies to spend \$1.2 billion on connectivity management solutions

(IDC, November 1, 2017)



The global 5G value chain will generate \$3.5 trillion in output and support 22 million jobs in 2035

(IHS Markit)

Sources

- <https://www.idc.com/research/viewtoc.jsp?containerId=US43161517>
- <https://www.gsmainelligence.com/research/?file=0efdd9e7b6eb1c4ad9aa5d4c0c971e62&download>
- <https://www.ihs.com/Info/0117/5g-technology-global-economy.html>
- <http://www.vodafone.com/business/iot/iotbarometer>

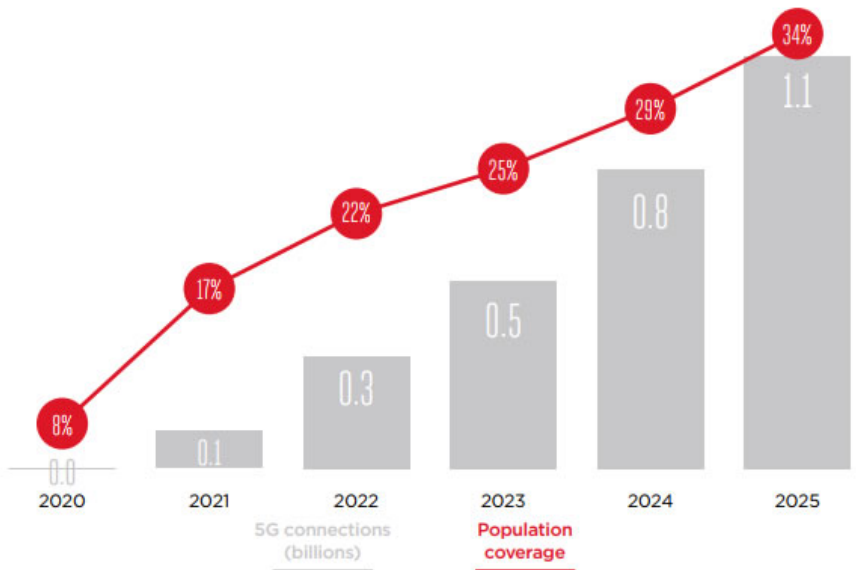
36% of organizations already think about 5G for IoT which they see as an incremental upgrade

(Vodafone IoT barometer, September 2017)

5G AND IOT

Source: GSMA Intelligence

Global 5G coverage and adoption

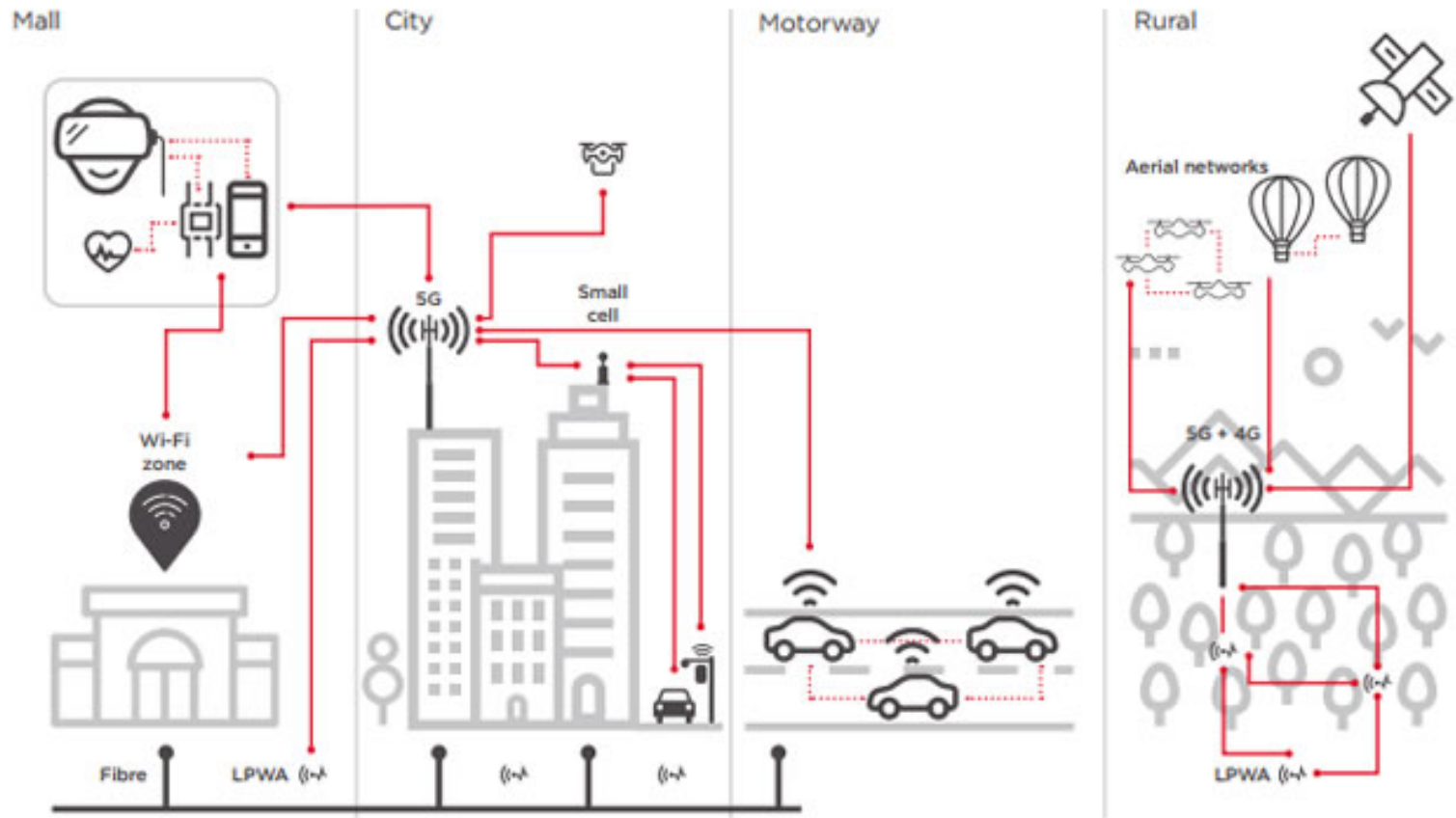


<https://www.i-scoop.eu/5g/5g-iot/>

5G AND IOT

Source: GSMA

5G as the centre of a heterogeneous network environment



<https://www.i-scoop.eu/5g/5g-iot/>

RADIO NETWORKS (RF LINKS)

- Simple radio communication interfaces
 - connecting to microcontrollers and devices via serial port
- Characteristics
 - Data rate: up to 1 Mbps
 - Range: up to 1 Km, depending on the antenna
 - Frequency: 2.4 GHz
- No direct support for high-level protocol such as TCP/IP
 - custom protocols

BLUETOOTH (BT)

- Standard widely adopted for transmitting data in Wireless Personal Area Network (WPAN), short range
 - **IEEE 802.15.1**
- Based on **dynamic discovery and connection** of devices in a short range (dozen meters)
 - widely used in the consumer devices (smartphones, tablets, laptops, printers, digital cameras...)
- Originally developed by Ericsson (1994) and then formalised as a standard by the Bluetooth Special Interest Group (SIG)
 - including Sony Ericsson, IBM, Intel, Toshiba, Nokia and others



BT: GENERAL INFO

- Standard designed with the main objective of low power consumption, short communication range and low cost for device production
- It works in the free 2,45 GHz bandwidth
 - to reduce interferences, the protocol divides the bandwidth in 79 channels and commute among channels 1.600 times per second (frequency hopping technique)
- Speed
 - versions 1.1 and 1.2 => up to 723,1 kbit/s.
 - versione 2.0 => high speed modality up to 3 Mbit/s
 - higher power consumption
- Newer versions adopt shorter signals and reduce the power required compared to Bluetooth 1.2

BT DEVICE CLASSES

- BT devices can belong to 4 different classes, depending on the power consumed and the working range

Classe	Potenza ERP		Distanza
	(mW)	(dBm)	(m)
1	100	20	~100
2	2,5	4	~10
3	1	0	~1
4	0,5	-3	~0,5

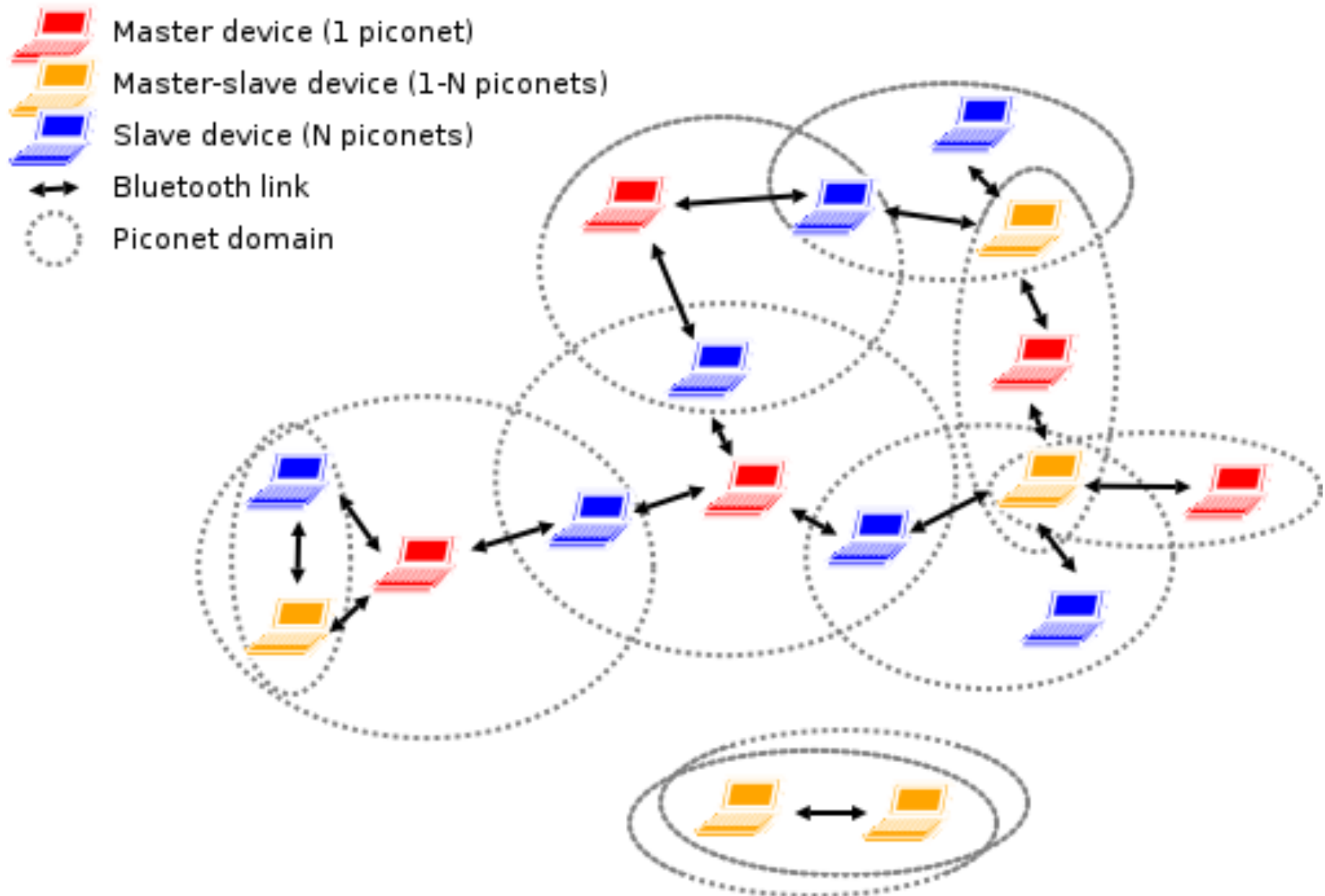
(Watt = Joule/sec, mW = milliWatt = 10^{-3} Watt)

(dB = decibel = $10 \cdot \log_{10}$ value)

BT NETWORK TOPOLOGY

- BT base network is called **piconet**, based on a **master-slave architecture**
 - each Bluetooth master device can manage the simultaneous communication with other 7 BT slave devices
 - a slave device can communicate only with one master at a time
 - the communication is synchronised with the master clock
- It is possible to connect multiple piconets, to form a **scatternet**.
 - slaves can take part to multiple piconets at the same time
 - a master of a piconet can be slave of another piconet.
- Every BT device can be configured to lookup and discover continuously other devices to connect
 - a password can be set for security

BT PICONET AND SCATTERNET

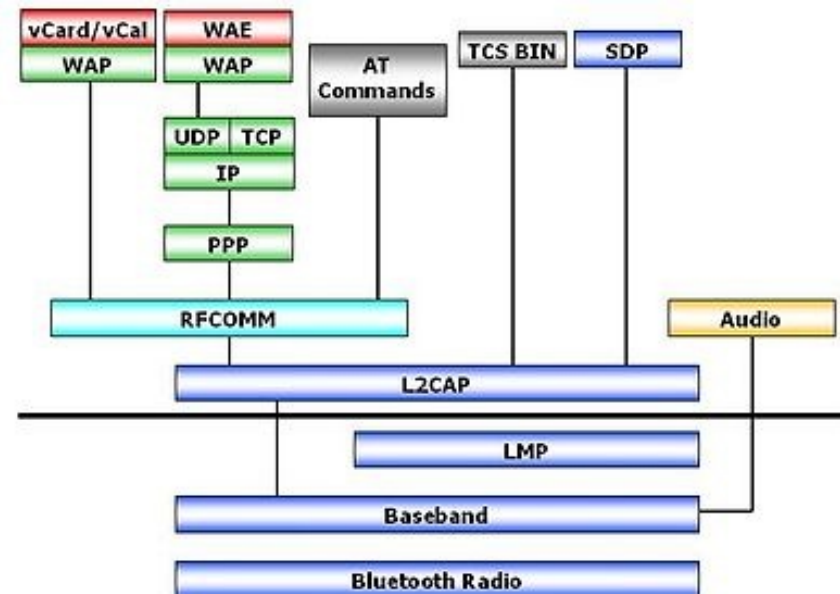
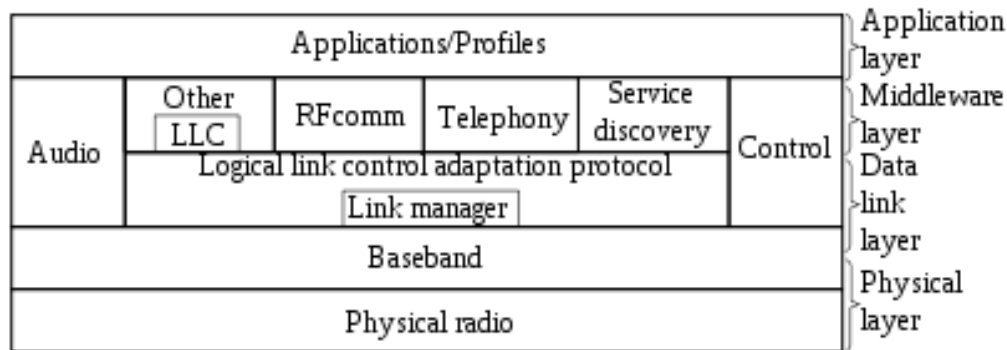


CONNECTION TYPES

- Two types in general
 - **connection oriented**
 - in this type first a connection must be established between devices before sending data
 - **connectionless**
 - a connection less link does not required a connection before sending packets
 - the transmitter can send packets anytime
 - it must know in any case the address of the target receiver
- Basic BT supports 2 connection types for voice and data transfer:
 - Asynchronous ConnectionLess (ACL)
 - for data transmission, best-effort
 - Synchronous Connection Oriented (SCO)
 - real-time transmission, multimedia

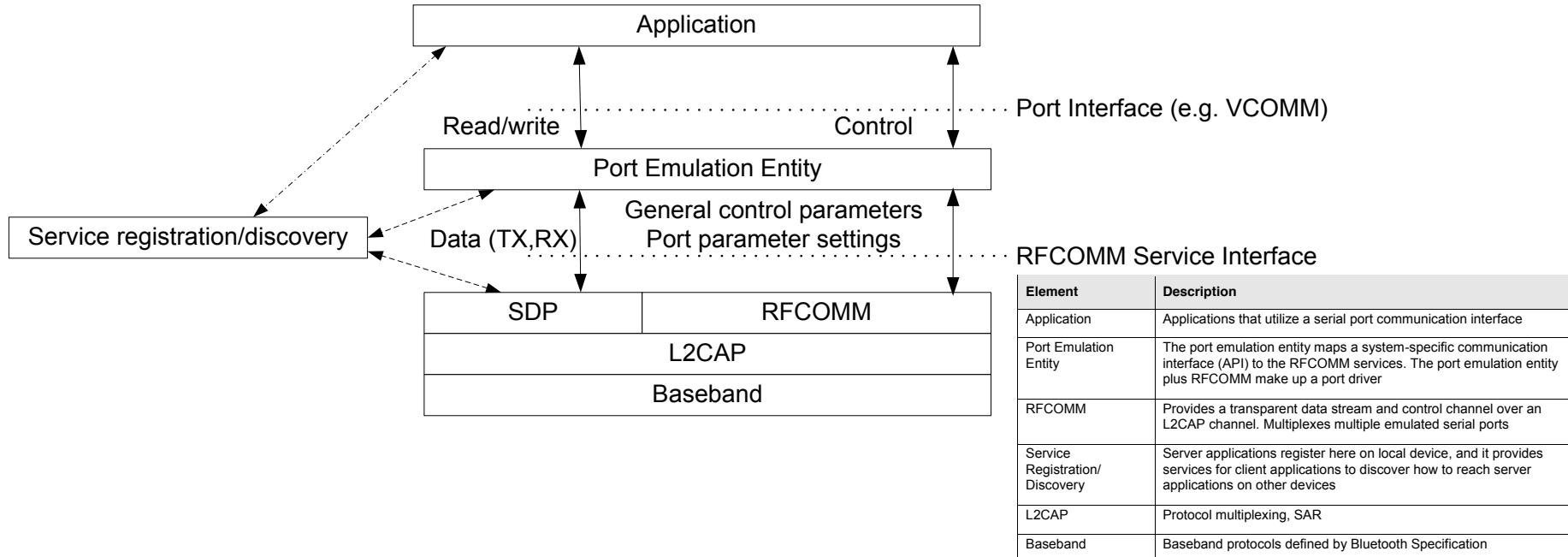
BT PROTOCOL STACK

- BT Protocol stack is organised in layers, like ISO/OSI
 - different protocols are used in different applications
 - the stack always include the data-link and physical layer



RFCOMM

- The RFCOMM protocol is used for exploiting BT for classic serial line communication, emulating the RS-232 serial port
 - useful for seamlessly connecting to any possible devices
- Reference model
 - <https://developer.bluetooth.org/TechnologyOverview/Pages/RFCOMM.aspx>



ADOPTED PROTOCOLS

- Protocols defined by different organisations for specific purposes
 - PPP
 - Internet standard for transmitting IP packets
 - TCP/UDP-IP
 - OBEX
 - object exchange, session protocol developed by Infrared Data Association, similar to HTTP
 - used to exchange data in vCard and vCalendar formats
 - WAE/WAP
 - Wireless Application Environment e Wireless Application Protocol

PROFILES

- To ease the adoption and use of BT devices, some specific **profiles** have been defined at the **application level**
- Examples
 - Generic Access Profile (GAP)
 - Service Discovery Application Profile (SDAP)
 - Cordless Telephony Profile (CTP)
 - Intercom Profile (IP)
 - Serial Port Profile (SPP)
 - Headset Profile (HSP)
 - Dial-up Networking Profile (DUNP)
 - Fax Profile
 - LAN Access Profile (LAP)
 - Generic Object Exchange Profile (GOEP)
 - Object Push Profile (OPP)
 - File Transfer Profile (FTP)
 - Synchronisation Profile (SP)
 - ...

SECURITY

- SAFER+ (Secure And Fast Encryption Routine) algorithm for authenticate devices and generate keys for data encryption

BT EVOLUTION AND BLE

- From 1.0 (1999) to version 5 (2016)
- Version 4.0 is called **Bluetooth Smart**, including protocols:
 - Classic Bluetooth - legacy BT
 - Bluetooth High Speed - based on Wi-Fi
 - **Bluetooth Low Energy (BLE)** - alternative protocol to BT specifically designed for IoT - Bluetooth 4
 - lower power consumption
 - easier communication among devices
 - higher reactivity
- Two implementations
 - *single-mode*
 - only the stack BLE is implemented
 - *dual-mode*
 - BT Smart functionalities are integrated with the Classic ones

CLASSIC BT vs. BT SMART

Technical Specification	Classic Bluetooth technology	Bluetooth Smart technology
Distance/Range (theoretical max.)	100 m (330 ft)	>100 m (>330 ft)
Over the air data rate	1–3 Mbit/s	1 Mbit/s
Application throughput	0.7–2.1 Mbit/s	0.27 Mbit/s
Active slaves	7	Not defined; implementation dependent
Security	56/128-bit and application layer user defined	128-bit AES with Counter Mode CBC-MAC and application layer user defined
Robustness	Adaptive fast frequency hopping, FEC , fast ACK	Adaptive frequency hopping, Lazy Acknowledgement, 24-bit CRC, 32-bit Message Integrity Check
Latency (from a non-connected state)	Typically 100 ms	6 ms
Minimum total time to send data (det.battery life)	100 ms	3 ms [31]
Voice capable	Yes	No
Network topology	Scatternet	Scatternet
Power consumption	1 W as the reference	0.01 to 0.5 W (depending on use case)
Peak current consumption	<30 mA	<15 mA
Service discovery	Yes	Yes
Profile concept	Yes	Yes
Primary use cases	Mobile phones, gaming, headsets, stereo audio streaming, smart homes, wearables, automotive, PCs, security, proximity, healthcare, sports & fitness, etc.	Mobile phones, gaming, smart homes, wearables, automotive, PCs, security, proximity, healthcare, sports & fitness, Industrial, etc.

BLUETOOTH 5

- Available from December 2016, this is the evolution of the standard toward IoT
 - 4x transmission area
 - at the same power consumption
 - 2x speed up to 2Mbps, in low power consumption mode
 - reduced time for sending and transmitting data

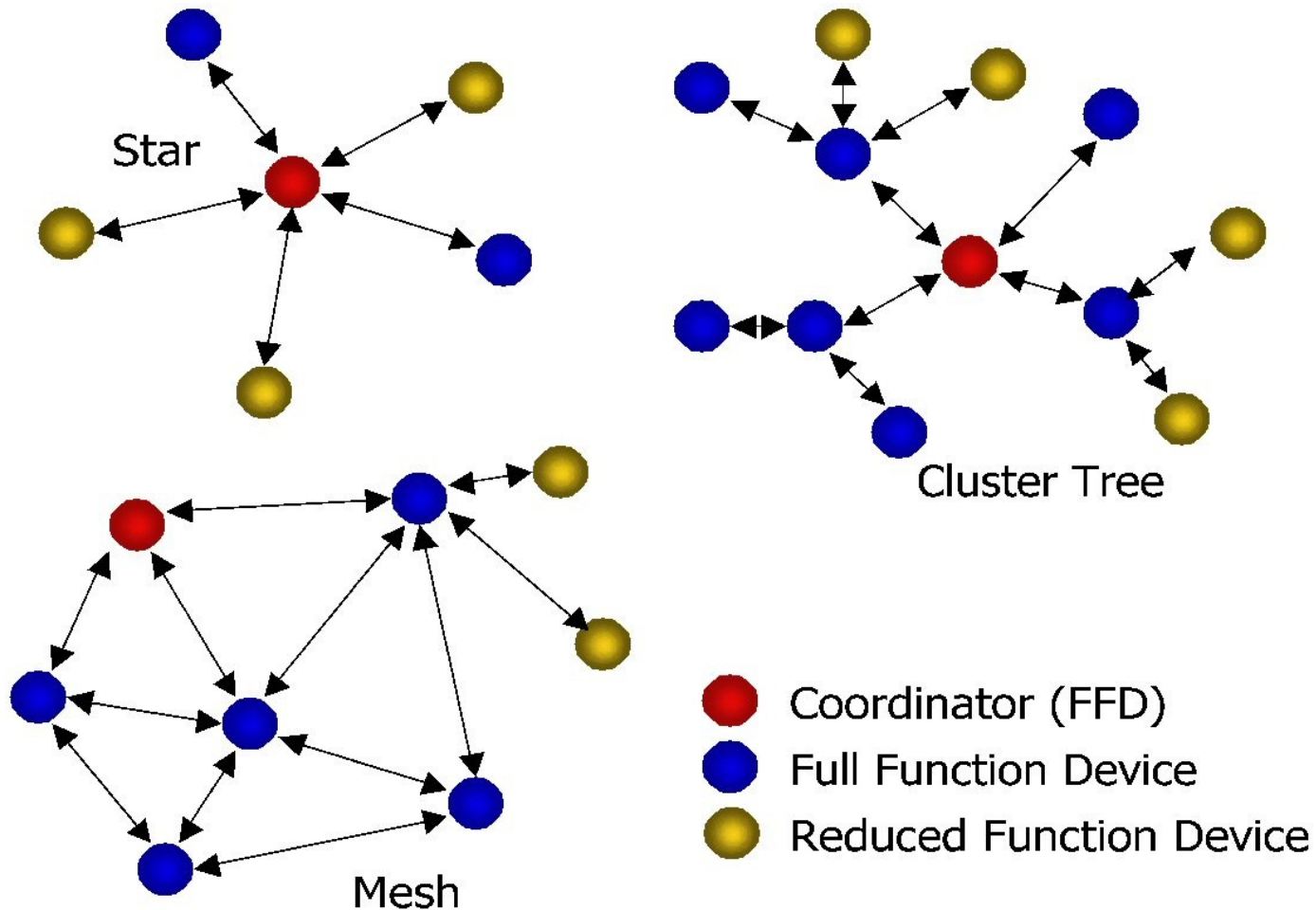
IEEE 802.15.4 STANDARD

- Standard that defines the specs for the physical layer and media access control for personal wireless network at low speed, low cost (*low-rate wireless personal area networks, LR-WPANs*)
 - focus on agile short-range communication among using low-power
 - within 10 meters, 250 kbit/s transfer rate
 - support for real-time communication and secure communication
- Base standard for further specs on top
 - **ZigBee**, ISA100.11a, WirelessHART, MiWi
- Used in:
 - Home and building automation
 - Automotive networks
 - Industrial wireless sensor networks
 - Interactive toys and remote controls

NETWORK MODEL

- Two node types
 - **FFD** (full-function device)
 - PAN **coordinator**
 - implements a communication model to talk with any other node
 - **RFD** (reduced-function devices).
 - these nodes represent simple devices — reduced computing power, reduced communication capabilities
 - can communicate with a FFD
- Topologies
 - **star e peer-to-peer** topologies
 - each network must have at least one FFD
 - **cluster tree**
 - structure in which a RFD can have a single FFD
- P2P network can establish arbitrary connection patterns
 - base for ad hoc networks for self-management and organization

NETWORK MODEL



CHARACTERISTICS

- Frequency
 - it works in 2.4 GHz free bandwidths, 900 MHz e 868 MHz
- Data rate
 - up to 250 kbps
- Range
 - from 100 m to 1 Km, depending on the transmitting power
 - small projects: 1mW => 100 m
- Consumption
 - 50mA
 - so a 850 mAH battery => 17 hours

IEEE 802.15.4: IMPLEMENTATIONS

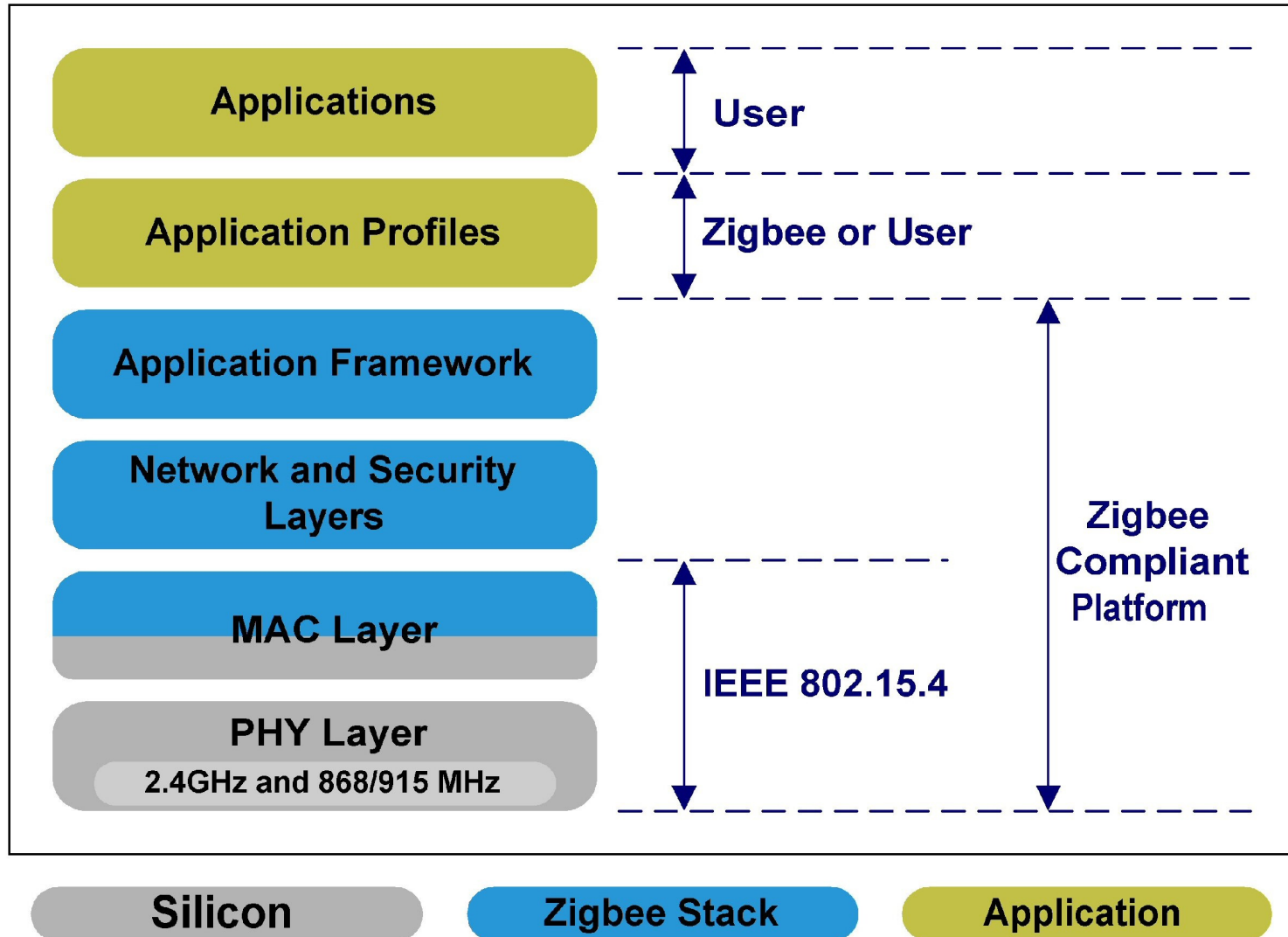
- Examples
 - ZigBee
 - 6LoWPAN
 - Thread

Protocol	Description
ZigBee	Promoted through the ZigBee Alliance, ZigBee defines upper-layer components (network through application) as well as application profiles. Common profiles include building automation, home automation, and healthcare. ZigBee also defines device object functions, such as device role, device discovery, network join, and security. For more information on ZigBee, see the ZigBee Alliance webpage, at www.zigbee.org . ZigBee is also discussed in more detail later in the next Section.
6LoWPAN	6LoWPAN is an IPv6 adaptation layer defined by the IETF 6LoWPAN working group that describes how to transport IPv6 packets over IEEE 802.15.4 layers. RFCs document header compression and IPv6 enhancements to cope with the specific details of IEEE 802.15.4. (For more information on 6LoWPAN, see Chapter 5.)
ZigBee IP	An evolution of the ZigBee protocol stack, ZigBee IP adopts the 6LoWPAN adaptation layer, IPv6 network layer, and RPL routing protocol. In addition, it offers improvements to IP security. ZigBee IP is discussed in more detail later in this chapter.
ISA100.11a	ISA100.11a is developed by the International Society of Automation (ISA) as “Wireless Systems for Industrial Automation: Process Control and Related Applications.” It is based on IEEE 802.15.4-2006, and specifications were published in 2010 and then as IEC 62734. The network and transport layers are based on IETF 6LoWPAN, IPv6, and UDP standards.
WirelessHART	WirelessHART, promoted by the HART Communication Foundation, is a protocol stack that offers a time-synchronized, self-organizing, and self-healing mesh architecture, leveraging IEEE 802.15.4-2006 over the 2.4 GHz frequency band. A good white paper on WirelessHART can be found at http://www.emerson.com/resource/blob/system-engineering-guidelines-iec-62591-wirelesshart--data-79900.pdf
Thread	Constructed on top of IETF 6LoWPAN/IPv6, Thread is a protocol stack for a secure and reliable mesh network to connect and control products in the home. Specifications are defined and published by the Thread Group at www.threadgroup.org .

ZIGBEE

- Widespread technology used in embedded systems
 - based on IEEE 802.15.4 standard, for wireless personal area networks (WPAN)
 - designed for IoT, for pervasive communication among devices
 - open global standard
 - low power consumption, low cost
- History
 - first studies in 1998, to overcome WiFi and Bluetooth limitations for future applications
 - the IEEE 802.15.4 standard was defined in 2003.
 - in 2005 the ZigBee Alliance produces the Specification 1.0

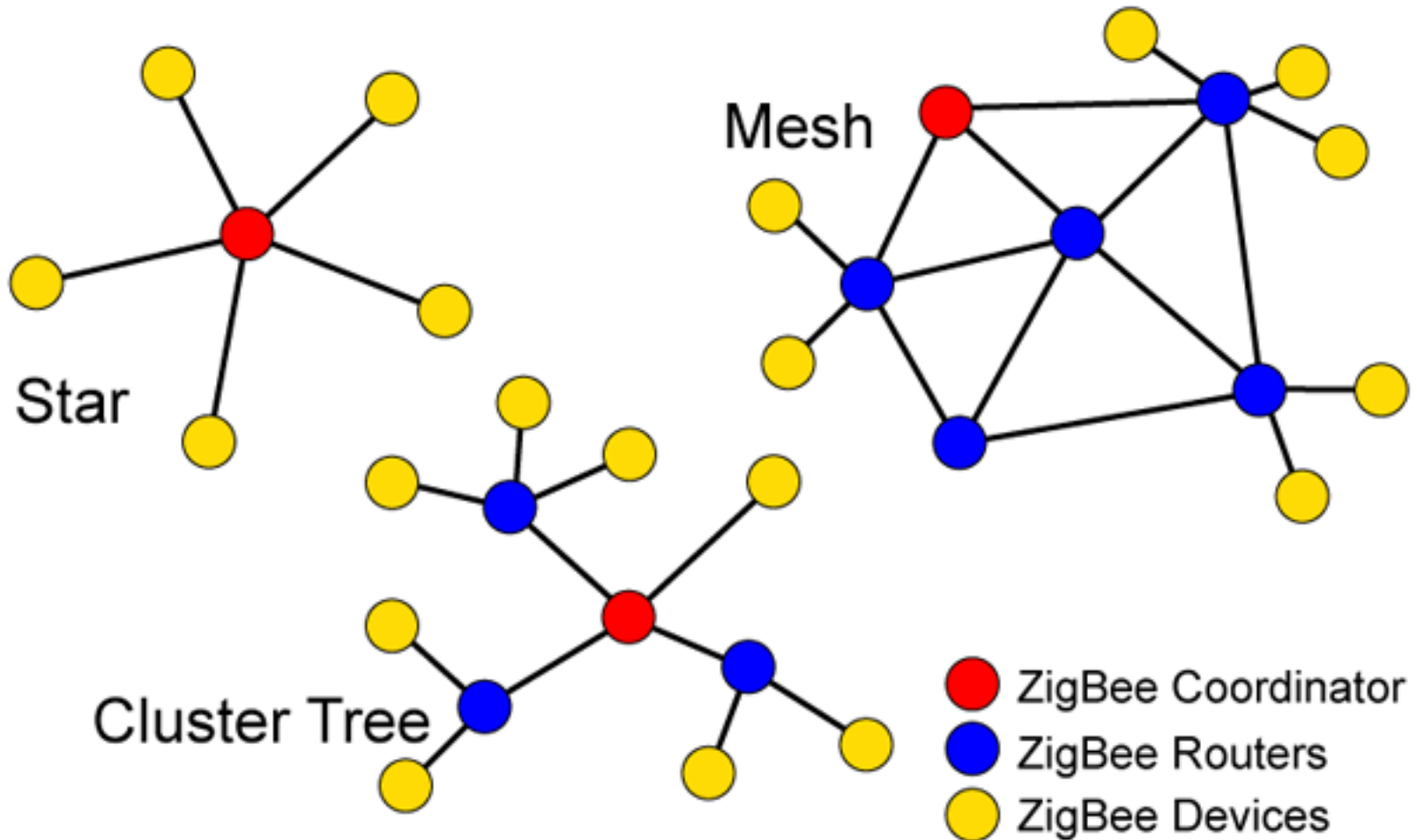
ZIGBEE STACK



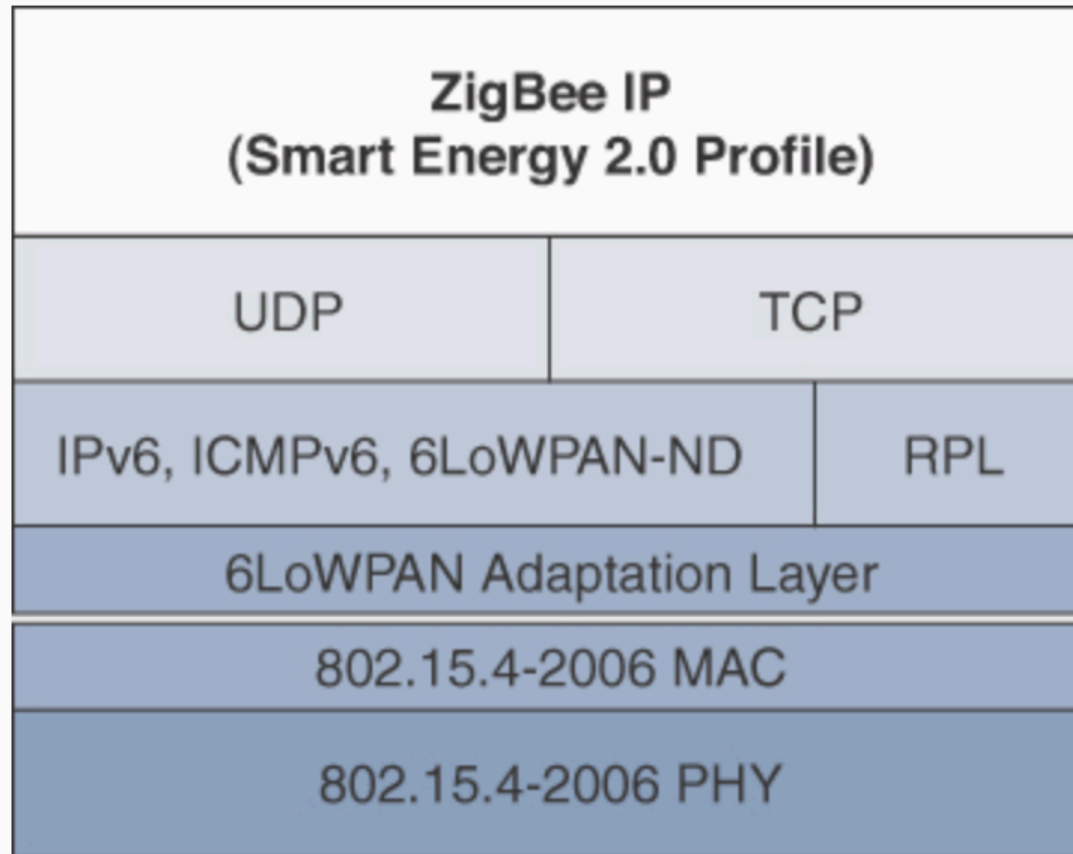
ZIGBEE ARCHITECTURE

- Three types of devices
 - **ZigBee Coordinator (ZC)**
 - it is the “intelligent” device among the devices, it is the root of the network and can function as a bridge
 - there could be only one Coordinator per network
 - can store info about the network and security management
 - **ZigBee Router (ZR)**
 - these devices act as intermediate routers transmitting data from/to other devices
 - **ZigBee End Device (ZED)**
 - these devices include only minimal functionalities for communicating with their parent (Coordinator or Router)
 - reduced memory/computing power

TOPOLOGIES



ZIGBEE IP STACK

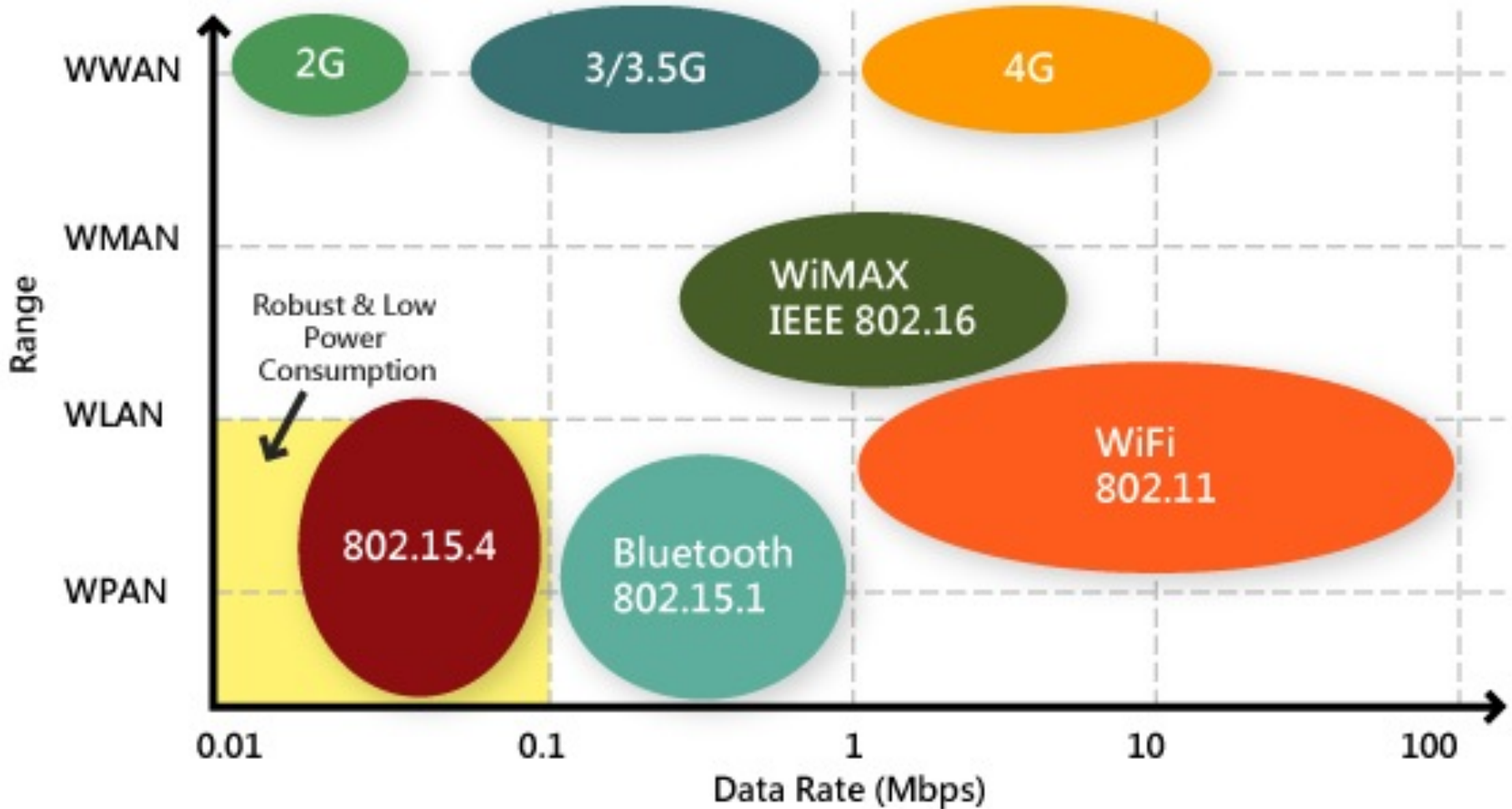


[IoT, p. 261]

ZIGBEE VS. BT

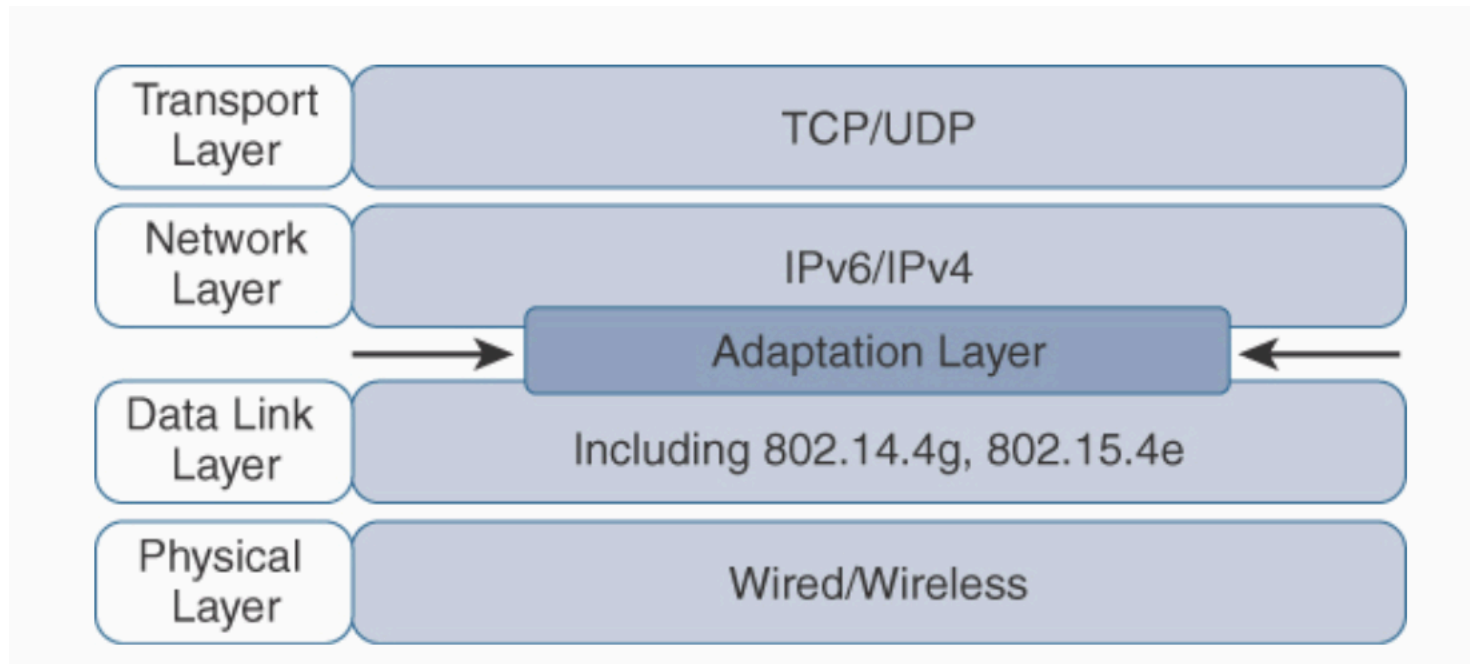
- ZigBee vs. classic BT
 - better efficiency, larger communication range, more agile creation of p2p networks
 - better for IoT
 - less used/widespread at the consumer level

COMPARISON



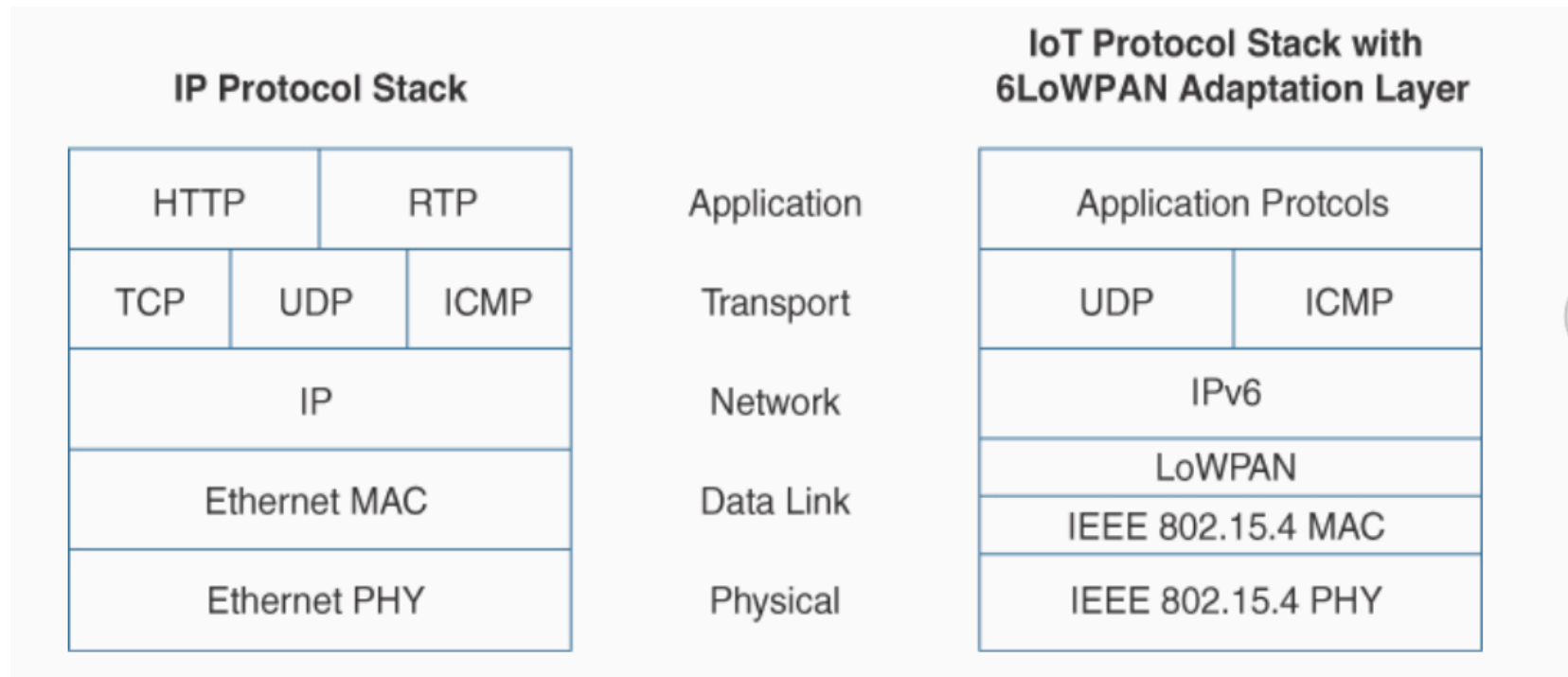
IP FOR IOT - OPTIMISATION

- IoT constrained devices/nodes define optimisations at different levels to work with the IP protocol/Architecture



EXAMPLE: 6LOWPAN

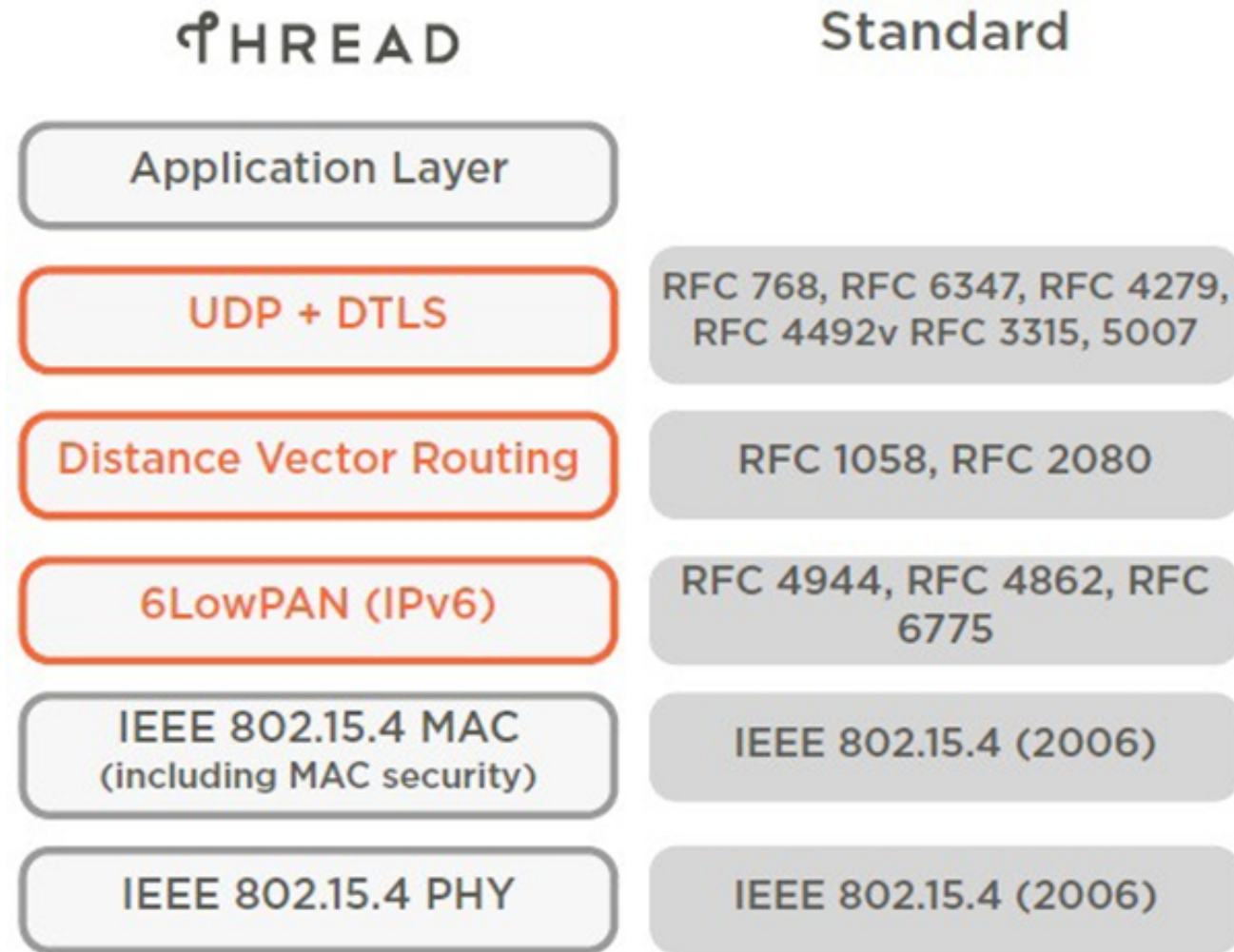
- Example: 6LoWPAN and 6Lo (successor of 6LoWPAN)
 - IPv6 packet optimisation on constrained network such as the IEEE 802.15.4



THREAD PROTOCOL

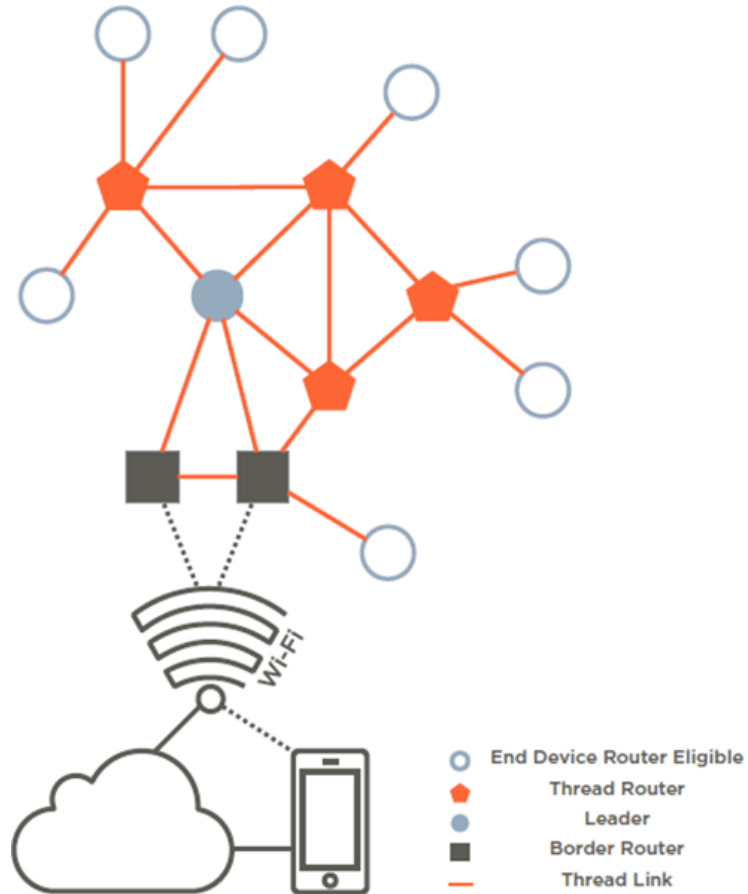
- IPv6-based networking protocol designed for low-power Internet of Things devices in an IEEE 802.15-2006 wireless mesh network, commonly called a Wireless Personal Area Network (WPAN)
- **Thread** is independent of other 802.15 mesh networking protocols, such a ZigBee, Z-Wave, and Bluetooth LE
 - especially designed for home and building automation, for for connecting products to each other, to the internet and to the cloud
- Thread's primary features include:
 - *Simplicity* — Simple installation, start up, and operation
 - *Security* — All devices in a Thread network are authenticated and all communications are encrypted
 - *Reliability* — Self-healing mesh networking, with no single point of failure, and spread-spectrum techniques to provide immunity to interference
 - *Efficiency* — Low-power Thread devices can sleep and operate on battery power for years
 - *Scalability* — Thread networks can scale up to hundreds of devices

THREAD NETWORK STACK



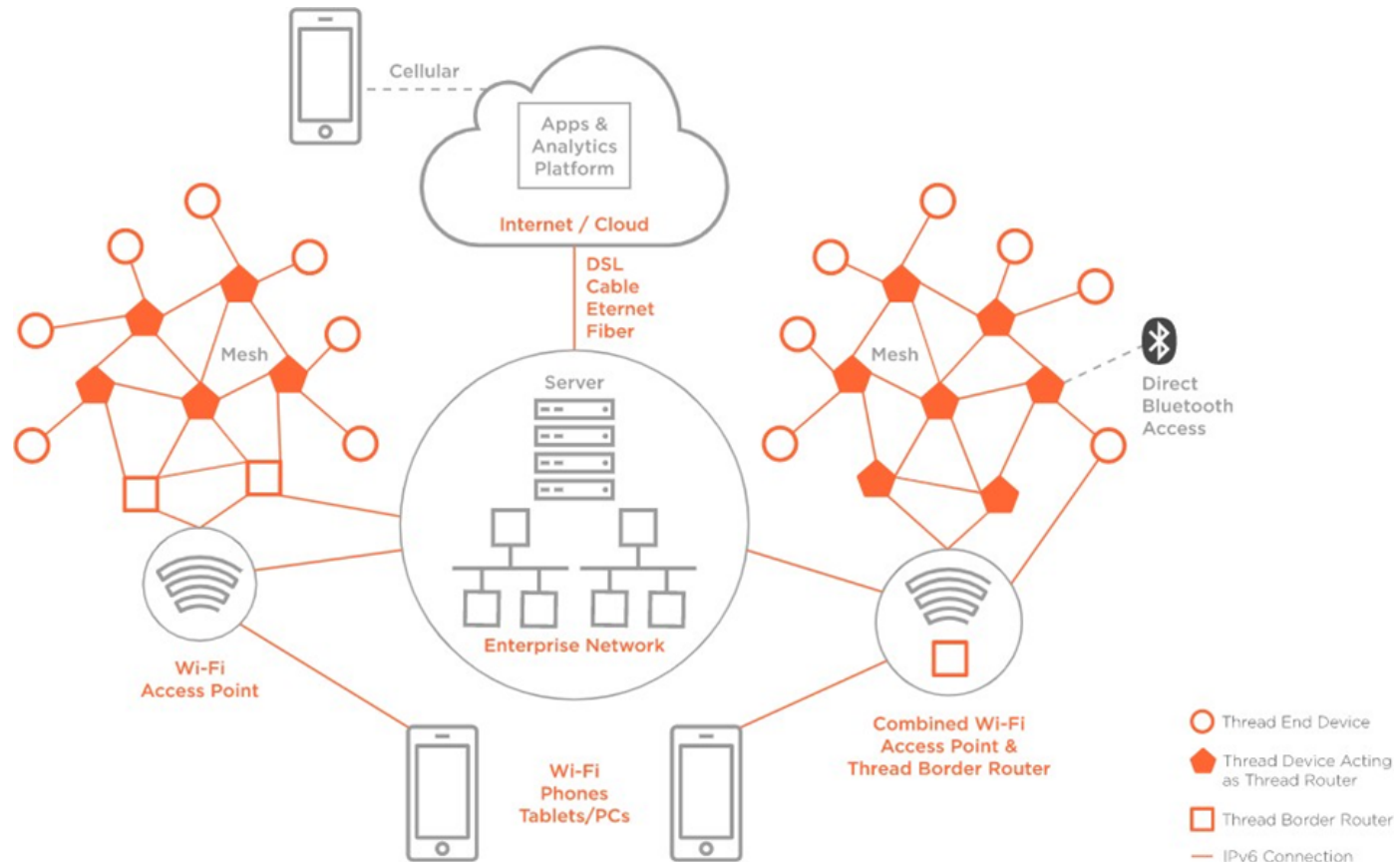
<https://raytac.blog/2023/07/12/thread-network-technology-and-key-feature-introduction/>

THREAD HOME NETWORK ARCHITECTURE



<https://raytac.blog/2023/07/12/thread-network-technology-and-key-feature-introduction/>

THREAD BUSINESS NETWORK ARCHITECTURE



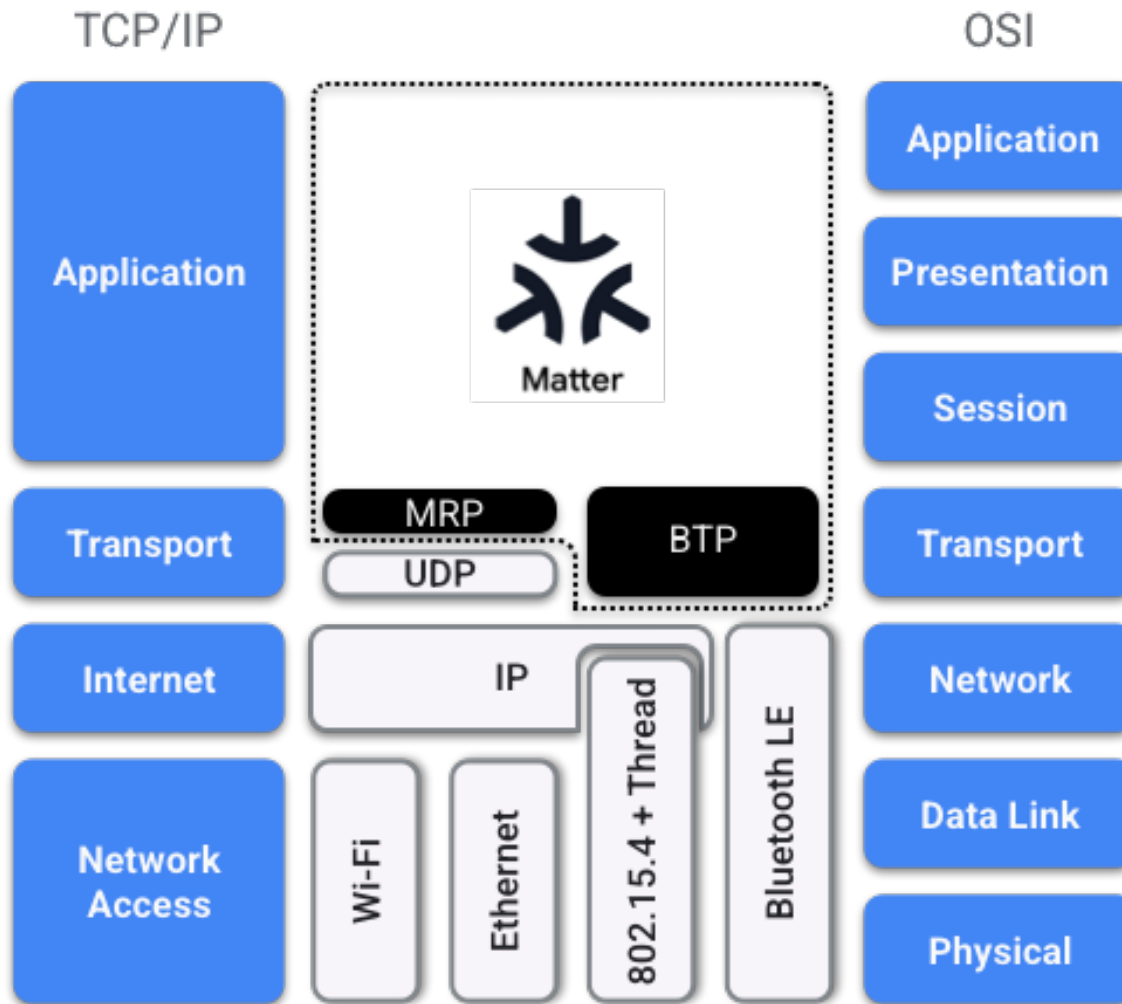
<https://raytac.blog/2023/07/12/thread-network-technology-and-key-feature-introduction/>

MATTER



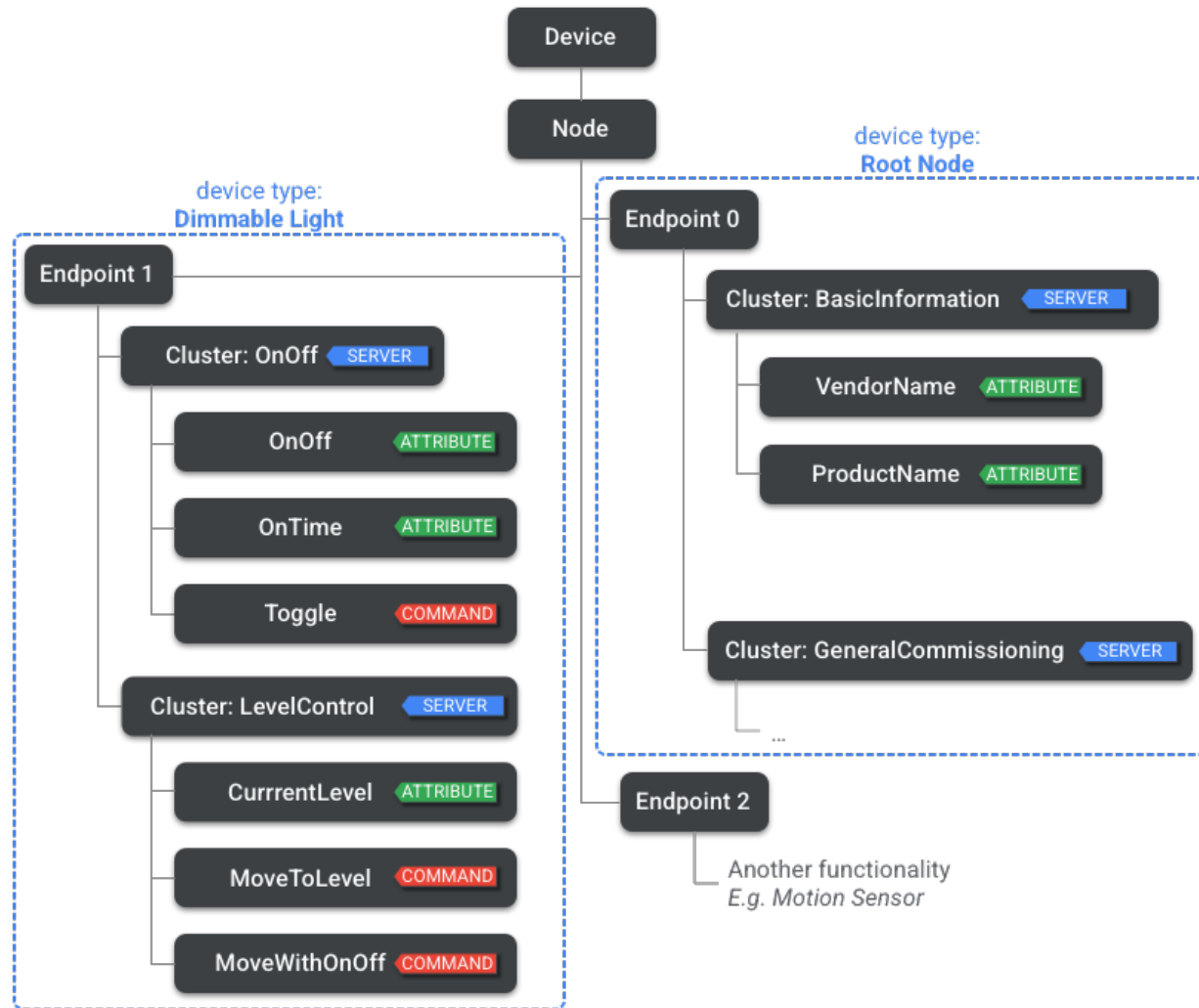
- Matter
 - Open-source connectivity standard for smart home and IoT (Internet of Things) devices
 - Recent initiative (2022)
 - strong, driven by major industries: Amazon, Apple, Google and the Connectivity Standards Alliance (formerly Zigbee Alliance)... followed by IKEA, Huawei, and Schneider
 - Royalty-free, but certification costs for developers and manufacturers
- Main focus
 - “Matter has the goal of being an **interoperable standard** that fosters technology adoption and innovation, gradually replacing proprietary protocols for smart home ecosystems” (<https://developers.home.google.com/matter/primer>)
 - Matter is implemented by an **open source SDK** that contains not only the implementation of the specification but also a rich set of examples and interoperable code

MATTER ARCHITECTURE



<https://developers.home.google.com/matter/primer/>

MATTER DEVICE MODEL



<https://developers.home.google.com/matter/primer/>

APPLICATION LEVEL PROTOCOLS

- Protocols for IoT — different possibilities and categories
- Application level protocols / categories
 - no application protocols (custom)
 - SCADA
 - web-based generic protocols
 - transport protocols like UDP/TCP
 - TCP to be preferred in cellular networks, that are typically more robust and can deal with the overhead
 - UDP to be preferred in LLN (*low-power and lossy*), characterised by constrained devices and networks
 - *application-level protocol designed for IoT (nodes and constrained networks)*
 - **MQTT** (Message Queueing Telemetry Transport)
 - **CoAP** (Constrained Application Protocol)

MQTT AND CoAP

CoAP	MQTT
UDP	TCP
IPv6	
6LoWPAN	
802.15.4 MAC	
802.15.4 PHY	

[IoTf, p. 414]

Example of stack
based on CoAP vs. MQTT

Factor	CoAP	MQTT
Main transport protocol	UDP	TCP
Typical messaging	Request/response	Publish/subscribe
Effectiveness in LLNs [*]	Excellent	Low/fair (Implementations pairing UDP with MQTT are better for LLNs.)
Security	DTLS	SSL/TLS
Communication model	One-to-one	many-to-many
Strengths	Lightweight and fast, with low overhead, and suitable for constrained networks; uses a RESTful model that is easy to code to; easy to parse and process for constrained devices; support for multicasting; asynchronous and synchronous messages	TCP and multiple QoS options provide robust communications; simple management and scalability using a broker architecture
Weaknesses	Not as reliable as TCP-based MQTT, so the application must ensure reliability.	Higher overhead for constrained devices and networks; TCP connections can drain low-power devices; no multicasting support

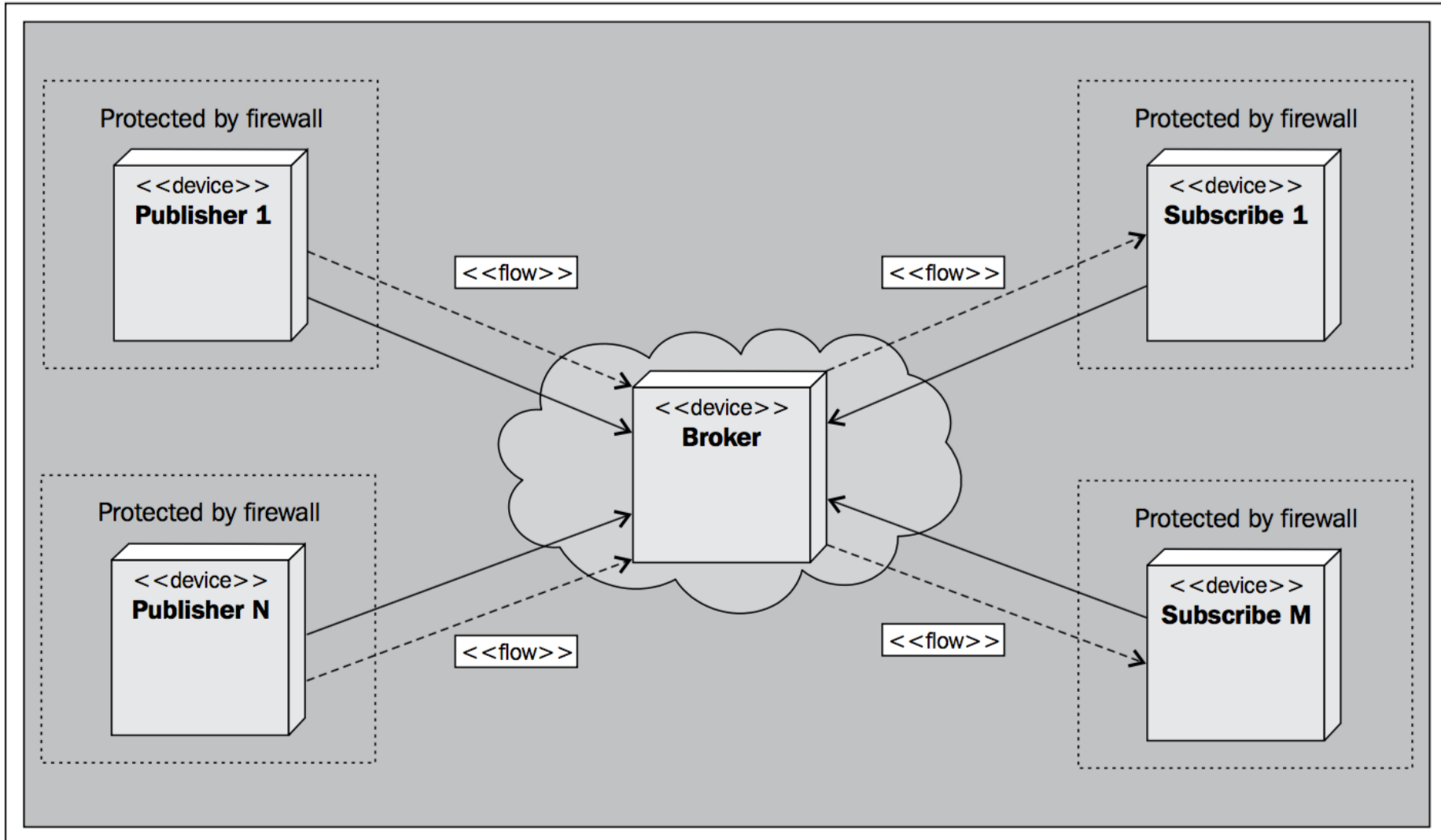
[*] LLN = low-power and lossy networks

[IoTf, p. 436]

PUBLISH SUBSCRIBE

- The MQTT protocol is based on the **publish/subscribe** pattern
- Three types of actors:
 - **Publishers**
 - they connect to the message broker and publish content
 - **Subscriber**
 - they connect to the same message broker and subscribe to content that they are interested in
 - **Message broker**
 - it makes sure that the published content is relayed to interested subscribers

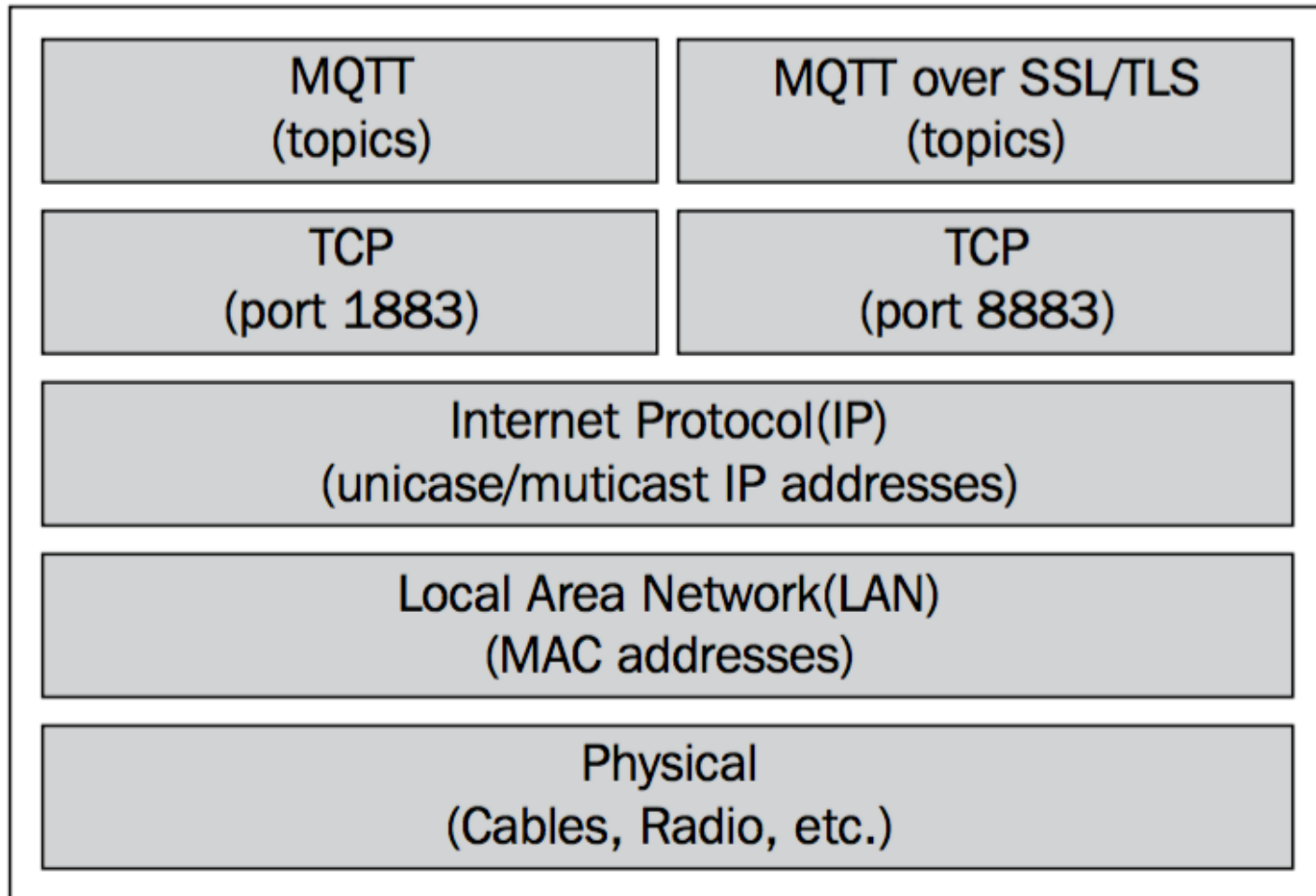
MQTT P/S ARCHITECTURE



TOPICS AND TOPICS TREE

- Content is identified by **topic**
 - when publishing content, the publisher can choose whether the content should be retained by the server or not
 - If retained, each subscriber will receive the latest published value directly when subscribing
- Topics are ordered into a **tree structure of topics**, much like a filesystem
 - the forward slash character (/) is used as a delimiter when describing a topic path.
 - when subscribing to content, a subscriber can subscribe to either a specific topic by providing its path, or an entire branch using the hash wildcard character (#)

MQTT PROTOCOL STACK



QUALITY OF SERVICE LEVELS

- There are three Quality of Service levels in MQTT available while publishing content.
 - **unacknowledged service**
 - the message is delivered *at most once* to each subscriber.
 - **acknowledged service**
 - each recipient acknowledges the receipt of the published information
 - If no receipt is received, the information can be sent again.
 - This makes sure the information is delivered *at least once*
 - **assured service**
 - the information is not only acknowledged but sent in two acknowledged steps
 - first transmitted and then delivered
 - this makes it possible to make sure that the content is delivered *exactly once* to each subscriber.

WEAK SECURITY

- Weak support for user authentication
 - plain text username and password authentication exists, but it provides an obvious risk if the server is not hosted in a controlled environment.
- To circumvent the most obvious problems, MQTT can be used over an encrypted connection using SSL/TLS.
 - in this case, it is important for clients to validate server certificates else user credentials may be compromised
- Other methods, not defined in the MQTT protocol itself, include the use of *client-side certificates* or *pre-shared keys* to identify clients, instead of using the username and password option provided by the protocol
- Proprietary methods of encrypting the contents can also be used to make sure only receivers with sufficient credentials can decrypt the contents.
 - even though this method works, it reduces interoperability and provides an additional load on each device, which is contrary to the stated goal of the protocol.
- As the MQTT protocol itself does not consider security, it is very important for developers to consider security themselves.

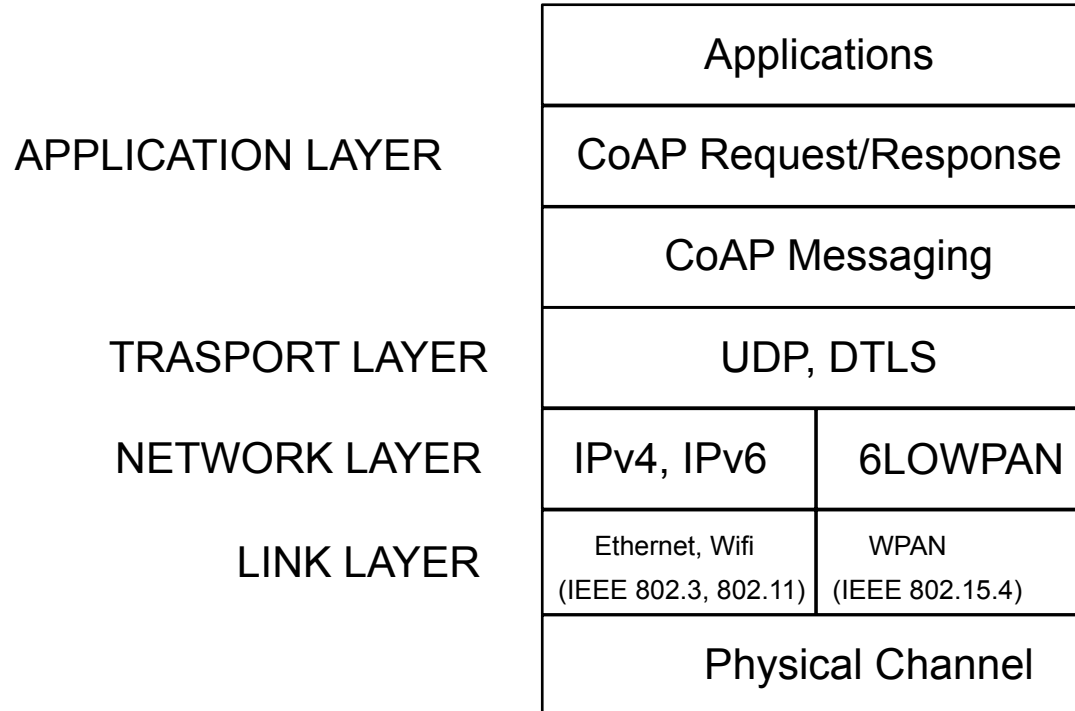
MQTT TECHNOLOGY

- Main reference
 - <http://mqtt.org>
 - OASIS standard:
 - <https://docs.oasis-open.org/mqtt/mqtt/v5.0/mqtt-v5.0.html>
- Available software
 - <https://mqtt.org/software/>
 - Two main kinds
 - clients - <https://github.com/mqtt/mqtt.org/wiki/libraries>
 - among the others:
 - » Arduino library - <https://github.com/knolleary/pubsubclient>
 - servers - <https://github.com/mqtt/mqtt.org/wiki/servers>
 - among the others:
 - » Eclipse Mosquitto - <https://www.mosquitto.org/>

CoAP

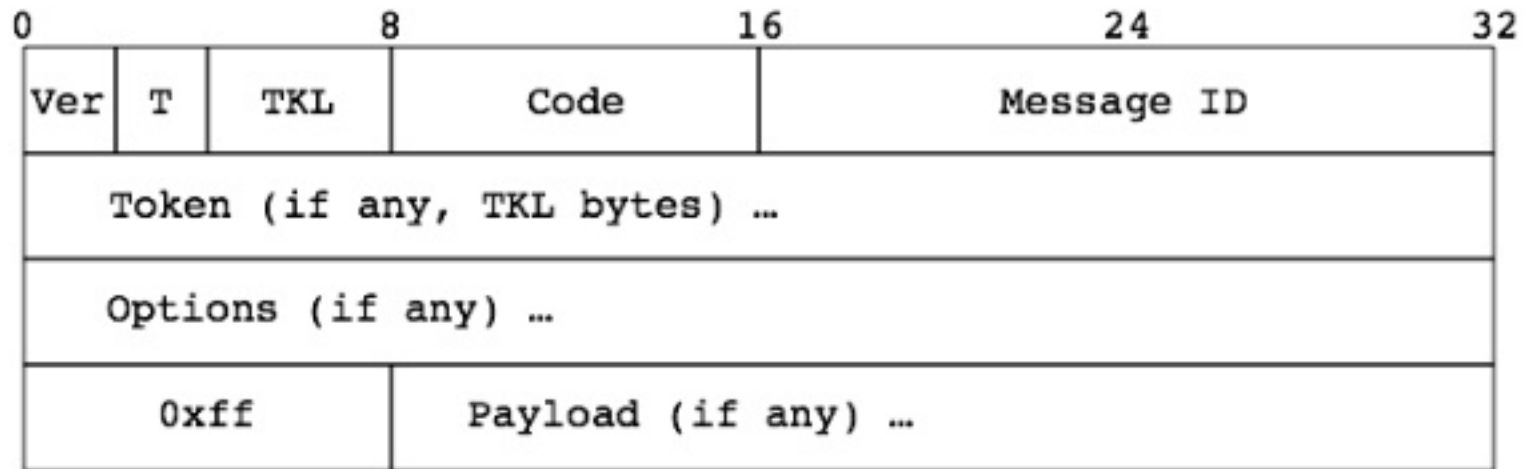
- Constrained Application Protocol (CoAP)
 - specialized Internet Application Protocol for constrained devices
 - RFC 7252: <https://tools.ietf.org/html/rfc7252>
 - It enables those constrained devices called "nodes" to communicate with the wider Internet using similar protocols
- CoAP is a service layer protocol
 - a main goals of CoAP is to design a generic web protocol for the special requirements of this constrained environment, especially considering energy, building automation, and other machine-to-machine (M2M) applications.
 - The goal of CoAP is not to blindly compress HTTP [RFC2616], but rather to realize a subset of REST common with HTTP but optimized for M2M applications.
 - can run on most devices that support UDP or a UDP analogue.

CoAP PROTOCOL STACK



CoAP MESSAGE FORMAT

- Minimum message size: 4 bytes (empty message)
- Fitting inside a UDP packet



SIMPLIFIED HTTP

- Implement a simplified HTTP
 - compact binary headers
 - reduces the number of options available in the header
- Reduced set of methods
 - GET, POST, PUT, and DELETE
 - *confirmable* and *nonconfirmable* message services
 - with a confirmable message, the receiver always returns an acknowledgement
 - the sender can, in turn, resend messages if an acknowledgement is not returned within the given time period
- Reduced number of response code
- Simplified Internet Media Type scheme => reduced set of Content-Formats
 - <http://www.iana.org/assignments/core-parameters/>

NEW FEATURES & EXTENSIONS

- New features wrt HTTP
 - CoAP supports multicasting.
 - this can be used to detect devices or communicate through firewalls
- Useful extensions
 - block transfer algorithm
 - to transfer larger amounts of data.
 - event subscription and notification architecture
 - observable resources emitting notifications when events occur
- Security
 - encryption based on the Datagram Transport Layer Security (DTLS)

TECHNOLOGY

- Overview
 - <http://coap.technology>
- Examples
 - **Californium**
 - <https://github.com/eclipse/californium>
 - Copper (Cu)
 - Firefox add-on, which allows to view and interact with CoAP resources
 - <https://addons.mozilla.org/en-US/firefox/addon/copper-270430>
 - A web-based CoAP test tool: <http://coap.me/>

MQTT

- One of the major problems we can encounter when using HTTP or CoAP protocols is *how to cross firewall boundaries*
 - firewalls not only block incoming connection attempts, but they also hide a home or office network behind a single IP address
 - unless the firewall blocks outgoing connections, which it does only if explicitly configured to do so, we can cross firewall boundaries if *all the endpoints in a conversation act as clients to a common message broker that lies outside of the firewall and is therefore accessible to everybody.*
 - the message broker acts as a server, but all it does is relay messages between clients.
- One protocol that uses message brokers is the **Message Queue Telemetry Transport (MQTT)** protocol
 - <http://mqtt.org/>
 - <https://mqtt.org/software/>

COMPARISON AT A GLANCE

Feature	HTTP	CoAP	MQTT
Request/Response	✓	✓	✗
Publish/Subscribe	✗	✗	✓
Multicast	✗	✓	✗
Events or Push	✓	✓	✓
Bypasses firewall	✗	(✓)	✓
Federation	✗	✗	✗
Authentication	✓	✓	✓
Network Identity	(✓)	(✓)	✗
Authorization	✗	✗	✗
Encryption	✓	✓	✓
End-to-end encryption	✗	✗	✗
Compression	✓	✗	✗
Streaming	✓	✗	✓
Reliable messaging	✗	✗	✓✓
Message Queues	✗	✗	✗

REFERENCES

- **[IoTf]** *IoT Fundamentals*. Hanes et al. CISCO Press. 2017
- *Building Internet of Things with Arduino*, Charalampos Doukas